

A structural proof of the Karpelevič theorem

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Abstract

We give a self-contained proof of the Karpelevič theorem by reducing an extremal invariant polygon to a cyclic product with an exact real phase. A minimum branching count organizes the contacts, and face persistence makes a projective deformation applicable without separate tower-height cases. Convexity determines the sharp radius; explicit stochastic realizations and an independent Farey comparison identify the complete boundary. The resulting scalar equation also gives a uniform relative asymptotic for the radial deficit, including points arbitrarily close to Farey endpoints. It yields an N^{-3} loss in badly approximable directions and a worst-direction loss of order N^{-2} for optimal polygonal gauges.

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1 The boundary theorem

Let Θ_n be the set of individual eigenvalues of real row-stochastic matrices of order n . Let F_n list the reduced fractions in $[0, 1]$ with denominator at most n , in increasing order, and put $F_n^+ = F_n \cap [0, 1/2]$. We prove the following form of the Karpelevič theorem [3, 4].

Theorem 1.1 (Karpelevič, in Ito form). *For $n \geq 4$, consecutive fractions $f < g$ in F_n^+ determine one upper boundary arc of Θ_n . Orient the pair so that the smaller denominator comes first: use $(p/q, r/s) = (f, g)$ and $\omega = z$ when the left denominator is smaller, and $(p/q, r/s) = (1 - g, 1 - f)$ and $\omega = \bar{z}$ otherwise. Thus $q < s$ and $rq - ps = 1$. With $m = \lfloor n/q \rfloor$, the arc is the branch of*

$$\omega^s(\omega^q - \beta)^m = (1 - \beta)^m \omega^{mq}, \quad 0 \leq \beta \leq 1, \quad (1.1)$$

joining the corresponding roots of unity. More explicitly, on an interior ray put $\vartheta = \arg \omega \in (0, 2\pi)$ and

$$A = q\vartheta - 2\pi p, \quad B = \frac{2\pi r - s\vartheta}{m}.$$

Its boundary radius is the unique $\rho \in (0, 1)$ solving

$$\rho^{s/m} \sin A + \rho^q \sin B = \sin(A + B).$$

These arcs, in increasing argument, form the upper boundary. The lower boundary is their complex conjugate, and Θ_n consists of all radial segments from zero to the boundary.

The unit-circle points are exactly the roots of unity of orders at most n . The small orders are

$$\Theta_1 = \{1\}, \quad \Theta_2 = [-1, 1], \quad \Theta_3 = \text{conv}\{1, e^{2\pi i/3}, e^{-2\pi i/3}\} \cup [-1, -1/2].$$

The order-three terminal upper boundary is the vertical segment from $e^{2\pi i/3}$ to $-1/2$, followed by $[-1, -1/2]$.

The intermediate product

The geometric part has a precise destination, stated fully in Theorem 4.1. Take a nonreal radial extremum inside the open unit disk, at its least matrix order $N \geq 4$. For one choice of conjugate multiplier $\omega = \rho e^{i\vartheta}$, a suitably chosen invariant polygon yields Farey neighbors $p/q < \vartheta/(2\pi) < r/s$, with $q < s$, and coefficients $0 \leq \beta_j < 1$. With $m = \lfloor N/q \rfloor$ and $e = s - mq$, they satisfy

$$\omega^e \prod_{j=1}^m (\omega^q - \beta_j) = \prod_{j=1}^m (1 - \beta_j), \quad e\vartheta + \sum_{j=1}^m u_j = 2\pi(r - mp).$$

Here u_j is the argument of the same factor $\omega^q - \beta_j$ in $[A, M)$, where $A = q\vartheta - 2\pi p$ and $M = \text{Arg}(\omega^q - 1) \in (A, \pi)$. If $e < 0$, the first identity is understood for $\omega \neq 0$, or after clearing the negative power. The second identity is an equality of real numbers, not just a congruence.

The half-open contact convention fixes these arguments, and the tower paths retain their total winding. Once both identities are available, the phase fixes the mean argument and the modulus fixes a sum of convex one-factor values. Equality of the coefficients then follows from the sharp comparison. The geometry therefore needs to produce one such optimized contact representation; it does not classify every invariant polygon at the extremal eigenvalue.

Relation to earlier work

Dmitriev and Dynkin obtained early restrictions on stochastic eigenvalues and determined the small orders [2]; Karpelevič described the region in every order [3]. Đoković later gave a simplification using cyclic polygons generated by powers of a complex number [24]. Ito's polynomial formulation [4] makes the boundary description especially concise. His argument establishes its equivalence with Karpelevič's theorem. A direct proof must instead explain why these polynomials describe the boundary, starting from the stochastic eigenvalue problem itself.

Realization, continuation, and outermostness are distinct parts of that task. Johnson and Paparella construct stochastic matrices realizing the roots of the relevant Ito polynomials [5, Theorem 3.2]. Munger, Nickerson, and Paparella organize a direct proof around three assertions: realization, a continuous selection joining the appropriate roots of unity through a Farey interval, and extremality of that selection on its rays [6, Proposition 1.2]. Kirkland, Laffey, and Šmigoc give a scalar trigonometric characterization of the unique boundary point on a prescribed ray, taking the Karpelevič boundary description as an input [20, Theorems 1.1 and 1.2]. In the present proof the product relation is derived from an arbitrary radial extremum. The convex comparison and an independent Farey refinement argument then identify its radius. The same scalar equation is recovered within the proof, before it is used for the later asymptotics.

The structure of realizing matrices has also been studied in greater detail than is needed for existence. Kirkland and Šmigoc classify realizations of the Type 0 and Type I reduced Ito polynomials and the sparsest realizations of Types II and III [21]. The subsequent preprints [35, 36] treat the non-degenerate full-degree Type II case and the Type III realization conjecture over its stated nonzero parameter range. These classifications concern matrices attached to prescribed boundary polynomials. Theorem 5.3 keeps the Johnson–Paparella graph and permits its return weights to vary independently. A cycle expansion computes the full characteristic polynomial. Appendix B records the graph's structural consequences and uses the convex comparison to identify the equal-weight tuple at the outer radius on the prescribed real phase. This last qualification is essential: the larger polynomial family also has roots on other phase branches.

The unrestricted region provides a reference point for structured stochastic families. Kirkland determined the non-Perron single-eigenvalue region of stochastic Leslie matrices in every order [19]. For matrices supported on a directed cycle and the diagonal, Ran and Teng formulated a four-dimensional conjecture [26, Conjecture 20]; the four-cycle region and its extension to every order are treated in [31, 34]. Work on monotone stochastic matrices determines the region through order three, provides realizations, and gives an all-order reduction [33]. Star-convexity for stochastic eigenvalue regions and several subclasses is studied in [32]. These are questions about the union of individual eigenvalues over a matrix family, which is different from prescribing the whole spectral list of one matrix.

Additional structural constraints also indicate where a further extension would require new arguments. The doubly stochastic single-eigenvalue problem belongs to the Perfect–Mirsky programme [25]. The proposed region is exact through order four [23], but fails in order five [17]. Doubly stochastic matrices, higher-order monotone matrices, and sparse support classes are therefore natural comparison problems for a theory based on phase-constrained products. The present paper proves no boundary classification for these additional families.

Proof strategy and consequences

The proof is self-contained apart from elementary linear algebra, the Perron–Frobenius theorem, and basic convexity and calculus. The segment comparison in Section 6 uses the one-variable fundamental theorem of calculus along a complex line segment; its application is to the polynomial z^m . Standard references for the background are Berman and Plemmons [9], Schneider [28], Coxeter [12], and Hardy and Wright [16]. We prove the invariant-polygon correspondence, the deflation and projective statements, the return arithmetic, and the stochastic realization used below. No part of the Karpelevič boundary theorem is used as an input.

The invariant-polygon correspondence and boundary-contact principle of Dmitriev and Dynkin [2], also developed in Swift’s account [30], turn the spectral question into one about a contracting rotation T . At a radial maximum of least order, every image vertex must meet the polygon boundary. Here this follows by removing a little stationary mass from a stochastic realization and normalizing its Perron eigenvector. We give the stationary-vector proof of this classical rank-one deflation principle [7, 8].

The geometric reduction follows Karpelevič’s architecture of vertex replacement, finite-rotation data, and a projective deformation forcing adjacent return [3]. Here it is governed by the number of vertex images that lie inside sides. That count cannot increase under an intermediate invariant hull with the same number of vertices. Exchanging two stochastic factors moves these contacts while preserving the spectrum; minimizing their number before area places them in one interval. The remaining obstruction is a first return that skips over a base vertex. A general fact about invariant polytopes—the dimension of the smallest face containing an orbit point cannot decrease—supplies the exposing lines. An explicit positive-denominator chart makes the transfer through the convex chain a positive affine recursion in reciprocal coordinates. Its closing comparison moves one image vertex strictly inward while preserving every other incidence, contradicting boundary contact.

The deterministic paths can now be contracted to a cyclic product. A cyclic recurrence with side-dependent coefficients already occurs in Karpelevič’s proof [3, Theorem VI and equation (13)]; he equalizes the coefficients geometrically before eliminating the vertices. We retain the unequal coefficients in the product together with their integer winding, since the product equation alone would not specify the boundary branch. After normalization, the factors lie on one affine line. The one-factor logarithmic convexity developed for the restricted n -cycle family [34, Lemmas 4.2 and 5.2] then gives the sharp scalar upper bound and forces equality of the coefficients at the boundary. A cycle expansion on the Johnson–Paparella

graph proves attainment. For the comparison between orders, one segment-integral identity handles both sides of a Farey median. This compares a least-order radial maximum with the candidate at the prescribed order.

The completed boundary theorem also solves an optimization problem for polygonal gauges. For a rotation followed by multiplication by ρ , the optimal gauge factor among polygons with at most N vertices is exactly $\rho/K_N(\theta)$. Section 7 proves attainment and the sharp worst-angle value. Section 8 derives the Farey asymptotic from the exact scalar equation, including relative control arbitrarily close to either endpoint. This gives the different vertex budgets for a fixed badly approximable direction and for a guarantee uniform in direction.

Two refinements of the factor argument are collected after these consequences. Appendix A quantifies the Jensen gap and treats unequal local exponents through a capped allocation of angular deficits. Appendix B develops the independent-parameter family already constructed in the main proof: it records irreducibility, primitivity, sparsity, and endpoint matrices. The convex comparison identifies the equal-parameter member that attains the outer radius on the prescribed real phase. These refinements are not prerequisites for the boundary proof.

This argument develops the geometric strategy and consequences of the supplied manuscript [1]. The structural reductions and the comparisons are proved here, so the paper can be read independently of that manuscript.

2 Saturation and cyclic contacts

For $n \geq 1$, let Θ_n be the union of the spectra of all real row-stochastic matrices of order n . This set is compact, conjugation invariant, and contained in the closed unit disk: the stochastic matrices form a compact set, their operator norm on ℓ^∞ is one, and the equation $\det(\lambda I - A) = 0$ is closed. The inclusions $\Theta_n \subseteq \Theta_{n+1}$ follow by adjoining an identity block.

The invariant-polygon correspondence of Dmitriev and Dynkin [2, §§1–2], presented in English by Swift [30], is

$$\lambda \in \Theta_n \iff \begin{array}{l} \lambda P \subseteq P \text{ for a non-singleton polytope } P \subset \mathbb{C} \\ \text{with at most } n \text{ vertices} \quad (n \geq 2). \end{array} \quad (2.1)$$

Indeed, from $Av = \lambda v$ take $P = \text{conv}\{v_1, \dots, v_n\}$. If it is a singleton then $\lambda = 1$, for which any segment works. Conversely, expressing each λv_i as a convex combination of the vertices gives a stochastic eigenpair, which can be padded with an identity block. For nonreal λ , such a polytope is a polygon: a segment would give a real invariant line. For nonreal λ with $0 < |\lambda| \leq 1$, one has $0 \in \text{int } P$. Iteration puts 0 in P when $|\lambda| < 1$; when $|\lambda| = 1$, the averages of a nonzero rotational orbit tend to zero and do the same. In either case, a supporting half-plane through 0 would then contain an entire rotational orbit of a nonzero vector. A finite such orbit sums to zero and an infinite one is dense in the circle, so neither fits in that half-plane.

The sets Θ_n , $n \geq 2$, are star-shaped about zero. If $Av = \lambda v$, $\lambda \neq 1$, choose a stationary probability row vector π^T . Then $\pi^T v = 0$ and

$$[tA + (1-t)\mathbf{1}\pi^T]v = t\lambda v \quad (0 \leq t \leq 1).$$

For $\lambda = 1$, the matrix with rows $(1, 0)$ and $(1-t, t)$ realizes t . Write $R_n(t)$ for the maximum radius on the ray of argument t .

For the geometric reduction, fix a nonreal multiplier $\zeta = \rho e^{i\theta}$, $0 < \rho < 1$, satisfying

$$\begin{aligned} \min\{|\text{ext } P| : \zeta P \subseteq P, \text{ int } P \neq \emptyset\} &= N, \\ \text{for every } t > 1, \quad t\zeta &\text{ has no invariant polygon} \\ &\text{with at most } N \text{ vertices.} \end{aligned} \quad (2.2)$$

Write $Tz = \zeta z$. These hypotheses hold at the first order of occurrence of any nonreal radial maximum inside the unit disk. A real planar map with nonreal eigenvalues is real-linearly conjugate to multiplication by either eigenvalue, so this entails no loss of generality.

Notation and orientation. The upper-half-plane point in the boundary theorem is z . The geometric argument may choose its conjugate: ζ denotes the multiplier after arranging the half-open contacts, and $\theta_\zeta \in (0, 2\pi)$ denotes its argument. The product theorem then chooses $\omega \in \{\zeta, \bar{\zeta}\}$ so that the smaller Farey denominator comes first; its argument is ϑ . Every conjugation replaces an argument by 2π minus that argument and reverses the vertex list. It reflects an ordered Farey pair (f, g) to $(1 - g, 1 - f)$. Section 5 returns to the upper ray $\theta = \arg z \in (0, \pi)$ and explicitly makes this final choice of ω .

The integer notation follows the successive reductions:

N, κ, δ	Least matrix order, cyclic contact shift, and $\delta = \gcd(N, \kappa)$.
φ	Number of interior contacts after they form one interval.
ν, q, h, Δ	In the arithmetic suspension, the number of bases, short return time, excess return time, and return displacement. For the contact towers $\nu = \varphi$; the product construction reapplies the lemma with a possibly larger set of bases.
ℓ	Length of the convex chain in the projective lemma.
q, s, m, e	In the product, the ordered Farey denominators, the factor count $m = \lfloor N/q \rfloor$, and $e = s - mq$. These data are specified anew in each return case.

2.1 Perron deflation forces boundary contact

A rank-one eigenvalue shift, due to Brauer [7] and developed further by Goddard [8], gives the saturation needed throughout the proof. The contact conclusions go back to Dmitriev and Dynkin [2, §§1–2]; the deflation argument below proves them in the hereditary form needed for later replacements. Subtracting a multiple of the stationary row leaves the other eigendirections unchanged while decreasing the Perron root. A positive diagonal normalization then moves those eigenvalues radially outward. A positive row provides enough room to make this subtraction without changing the support.

Lemma 2.1 (Positive Perron deflation). *Let A be irreducible stochastic with stationary probability vector π^T . If $u \geq 0$, $B = A - u\pi^T$ is nonnegative and irreducible, and $r = 1 - \pi^T u \in (0, 1)$, then a positive diagonal similarity makes B/r stochastic. Its non-Perron eigenvalues are those of A divided by r , including algebraic multiplicities. Such a deflation exists whenever A has a positive row.*

Proof. The identity $\pi^T B = r\pi^T$, with $\pi > 0$, identifies r as the Perron root of B . For $Bh = rh$, $h > 0$, put $D = \text{diag}(h)$; then $r^{-1}D^{-1}BD\mathbf{1} = \mathbf{1}$. On the invariant hyperplane $\ker \pi^T$, B agrees with A , whereas its quotient eigenvalue is r . This proves the spectral assertion, and an eigenvector v becomes $D^{-1}v$. If row i is positive, take $u = \varepsilon e_i$ with $0 < \varepsilon < \min_j (a_{ij}/\pi_j) \leq 1$; B retains the support of A and $r \in (0, 1)$. $\square$

Theorem 2.2 (Hereditary saturation). *Under (2.2), every invariant polygon with at most N vertices has exactly N vertices; every image vertex belongs to its boundary; and every side meets its image.*

Proof. Let $v_1, \dots, v_N$ be the vertices of such a polygon and choose a stochastic realization $Av = \zeta v$ by barycentric coordinates. Every such A is irreducible: a proper nonempty closed

row set would restrict this equation to a smaller stochastic realization, whose eigenvector is nonzero because every polygon vertex is nonzero. That contradicts minimal order.

If ζv_i were interior, its barycentric coordinates could all be chosen positive, making row i positive. To see this, first assign a small equal weight to every vertex; the remaining point is still in the polygon. Lemma 2.1 would then realize ζ/r stochastically at order N , with $r < 1$, contradicting radial maximality and (2.1). Thus every image vertex is on the boundary.

The polar polygon $P^\circ = \{a : \langle a, x \rangle \leq 1 \text{ for all } x \in P\}$ has one vertex for each side of P . It is invariant under the adjoint map, since $\langle T^*a, x \rangle = \langle a, Tx \rangle \leq 1$; that map satisfies the same hypotheses by reflection. Apply the vertex-contact conclusion to it. For a polar vertex a , the boundary point T^*a satisfies $\langle T^*a, x \rangle = 1$ for some $x \in P$. Hence $\langle a, Tx \rangle = 1$, so the side of P exposed by a meets TP . $\square$

We use the following elementary face rule: if a positive convex combination of points of a polytope lies on its boundary, every point used lies in one common proper face. A supporting functional proves this by equality in its upper bound. For a polygon, positive support at a saturated contact therefore consists of at most two adjacent vertices.

2.2 Branching monotonicity

The number of vertices is fixed by minimal order. A second integer makes clipping monotone. For an injective linear map T and a T -invariant polytope P with vertex set V , define

$$d_T(P) = |\{v \in V : Tv \in V\}|, \quad b_T(P) = |V| - d_T(P).$$

The first counts deterministic contacts; the second counts images that are not vertices. In a saturated planar realization, these are precisely the images inside edges.

Theorem 2.3 (Monotonicity under intermediate hulls). *In any dimension, suppose*

$$TP \subseteq Q \subseteq P, \quad |\text{ext } Q| = |\text{ext } P|, \quad \text{ext } Q \subseteq \text{ext } P \cup T \text{ext } P.$$

Then Q is invariant and $b_T(Q) \leq b_T(P)$.

Proof. Put $W = \text{ext } Q$, $R = V \setminus W$, $A = W \setminus V$, and $E = \{v \in V : Tv \in V\}$. An extreme point of P lying in Q remains extreme in Q . Hence $R \cap TV = \emptyset$. Every old deterministic source in $E \setminus R$ therefore remains deterministic. Since $A \subset TV$, injectivity makes $S = T^{-1}A$ a subset of V of size $|A| = |R|$, disjoint from E . The retained sources $S \setminus R$ are new deterministic sources for Q , disjoint from the old ones. Consequently

$$d_T(Q) \geq |E| - |E \cap R| + |R| - |S \cap R| \geq |E| - |E \cap R| + |R| - |R \setminus E| = |E|.$$

Finally $TQ \subseteq TP \subseteq Q$. $\square$

Normalize the invariant N -gons in (2.2) by

$$\max_{x \in P} |x| = 1.$$

The normalized class is compact. Indeed, from any sequence choose a convergent subsequence of the N -tuples of vertices in the closed unit disk. Their limiting convex hull remains invariant, contains zero, and has radius one. The last two properties exclude a point. Nonreality of ζ excludes a segment, and minimal order excludes a polygon with fewer than N vertices. Thus all N limiting points are distinct vertices.

In these vertex coordinates, an equality $Tv_i = v_j$ can appear in a limit but cannot disappear. The count d_T is therefore upper semicontinuous, and b_T is lower semicontinuous. Its

minimum is attained on a compact set; continuity of area then gives an area minimizer within that set. Choose such a polygon P throughout this section: it minimizes $(b_T(P), \text{area}(P))$ lexicographically in the normalized class, meaning first branching, then area. Scaling changes neither count, so its branching count is minimal among all invariant N -gons, with or without the normalization.

These two choices have different uses. For an intermediate hull with N vertices drawn from $\text{ext } P \cup T \text{ ext } P$, Theorem 2.3 preserves minimum branching. If that hull also retains a radius-one vertex, it is still normalized; a proper inclusion would then contradict the area choice. Later interchanges will retain minimum branching even when they do not retain the area choice.

2.3 The cyclic contact representation

Write $[a, b]_{\partial P}$ for the boundary arc from a to b in the positive direction, with the indicated endpoints included or excluded. The half-open sides $(v_{i-1}, v_i]$ partition ∂P : each vertex belongs to the side ending there. We must choose an orientation in which these sides receive exactly one image vertex each. Boundary contact alone does not yet specify which side receives a contact at a vertex.

List the vertices y_j of TP in cyclic order. Saturation puts them on ∂P , in the same order, and ensures that every side of P meets TP . For the image edge $[y_j, y_{j+1}]$, intersect P with the closed half-plane bounded by its line and containing TP ; call the intersection Q_j . Let

$$k_j = |\text{ext } P \cap [y_j, y_{j+1}]_{\partial P}|.$$

If the cut is proper, its line meets the interior of P . The two boundary intersections are therefore exactly y_j, y_{j+1} , and the discarded boundary is the positive open arc between them. The cut removes the old vertices on that open arc and can add only its two endpoints. Counting both endpoints in k_j gives

$$TP \subseteq Q_j \subsetneq P, \quad |\text{ext } Q_j| \leq N + 2 - k_j.$$

The intermediate hull is invariant, so minimal order forces $k_j \leq 2$. When $k_j = 2$, it has exactly N vertices and all its new vertices lie in $T \text{ ext } P$. Theorem 2.3 therefore puts it in the same minimum-branching class.

Fix a vertex v_* of P with $|v_*| = 1$. Every y_j has radius at most $\rho < 1$, so v_* lies on exactly one of the disjoint open arcs $(y_j, y_{j+1})_{\partial P}$. A proper cut with $k_j = 2$ must discard v_* : otherwise Q_j would remain normalized and branch-minimal, but have strictly smaller area than P . Hence at most one proper cut has $k_j = 2$. If $Q_j = P$, the image-edge line supports P , and the arc lies on one side; its positive half-open part contains at most one old vertex.

The closed-arc bound controls the cuts. To determine endpoint assignments, set

$$r_j = |\text{ext } P \cap (y_j, y_{j+1})_{\partial P}|, \quad \varepsilon_j = \mathbf{1}_{\{y_j \in \text{ext } P\}}.$$

These half-open arcs partition the old vertex set. Consequently

$$\sum_j r_j = N, \quad k_j = r_j + \varepsilon_j, \quad 0 \leq r_j \leq 2,$$

and at most one r_j equals two. Either every $r_j = 1$, or there are unique indices s, t with

$$r_s = 2, \quad r_t = 0, \quad r_j = 1 \quad (j \notin \{s, t\}).$$

Consider the latter case. Since $k_s \leq 2$, necessarily $\varepsilon_s = 0$. Also $\varepsilon_{s+1} = 1$. Otherwise the two old vertices counted by r_s would both lie strictly between y_s and y_{s+1} . They are consecutive

vertices of P , and their joining side contains no image vertex. It could not meet TP : a supporting functional attaining its maximum on TP attains it at an image vertex. This contradicts side contact. Thus s is a transition from zero to one in the cyclic sequence (ε_j) .

A transition from one to zero can occur only at t . Indeed, suppose $\varepsilon_j = 1$ and $\varepsilon_{j+1} = 0$ with $j \neq t$. Then $j \neq s$, so $r_j = 1$ and $k_j = 2$. This cut must be proper. If its line supported P , the arc would start at the old vertex y_j and end at the nonvertex y_{j+1} on that side, before the next old vertex; it would then have $r_j = 0$. The proper cut at j would therefore be a second one with $k_j = 2$, a contradiction.

A cyclic binary sequence has equally many transitions in the two directions. The upward transition at s and the only possible downward transition at t must therefore be its only transitions. Switching which endpoint belongs to an arc gives

$$|\text{ext } P \cap [y_j, y_{j+1})_{\partial P}| = r_j + \varepsilon_j - \varepsilon_{j+1} = 1 \quad \text{for every } j.$$

Thus one of the two half-open conventions interlaces the vertex lists exactly. If each $[y_j, y_{j+1})_{\partial P}$ contains the old vertex u_j , then $y_j \in (u_{j-1}, u_j]$. If instead each $(y_j, y_{j+1}]_{\partial P}$ contains u_j , then $y_{j+1} \in [u_j, u_{j+1})$. In this second case replace P by $\bar{P}$ and ζ by $\bar{\zeta}$, and list the reflected vertices in positive cyclic order. This converts the assignment into sides open at their initial endpoint. Conjugation preserves the extremality hypotheses (2.2), the branching count, normalization, and area. Retain the notation P, ζ for the resulting polygon and multiplier.

Multiplication by ζ preserves cyclic order. The resulting bijection from source vertices to half-open target sides is therefore a translation of cyclic index sets. For a uniquely determined $1 \leq \kappa \leq N$,

$$c_i = \zeta v_{i-\kappa} = \beta_i v_{i-1} + \alpha_i v_i \in (v_{i-1}, v_i], \quad \alpha_i = 1 - \beta_i > 0, \quad (2.3)$$

with indices modulo N . If $\kappa = N$, every contact is interior, and conjugating with reversed vertex order gives the same representation with $\kappa = 1$ and interchanged coefficients. We henceforth arrange $1 \leq \kappa < N$.

Choose vertex arguments Φ_i increasing with i , extended by $\Phi_{i+N} = \Phi_i + 2\pi$. Since 0 is interior, each gap $\gamma_i = \Phi_i - \Phi_{i-1}$ lies in $(0, \pi)$. Polar argument increases along each side, so, for the representative $0 < \theta_\zeta < 2\pi$ of $\arg \zeta$,

$$\Phi_{i-1} < \Phi_{i-\kappa} + \theta_\zeta \leq \Phi_i, \quad \frac{\kappa - 1}{N} < \frac{\theta_\zeta}{2\pi} < \frac{\kappa}{N}. \quad (2.4)$$

There is no hidden multiple of 2π in the first inequality, because both its interval after subtracting $\Phi_{i-\kappa}$ and θ_ζ lie between zero and 2π . Summing gives the second assertion; equality on the right would make every contact deterministic and force a nonzero vertex to be fixed by a positive power of ζ , contrary to $|\zeta| < 1$.

2.4 The local operation and the global normal form

The local replacement exchanges two stochastic factors. If $A = HS$ with H, S stochastic and S invertible, then

$$A' = SH = SAS^{-1}$$

is stochastic with the same spectrum, and an eigenvector v becomes Sv . We choose S to move one polygon vertex into its preceding side.

Let C act on lists by $(Cv)_r = v_{r+1}$, and let $B(\beta)$ have row i equal to $\beta_i e_{i-1}^T + \alpha_i e_i^T$. Then (2.3) is the stochastic eigenpair

$$A = C^\kappa B(\beta), \quad Av = \zeta v.$$

For the deterministic idempotent E_i that sends coordinate i to $i - 1$ and fixes the others, put $S_i(t) = (1 - t)I + tE_i$.

The coefficients are most transparent as nonnegative weights $w_i = -\log \alpha_i$. The update below transports one such weight along the cyclic shift and adds it to the weight at its destination:

$$w'_i = 0, \quad w'_{i+\kappa} = w_i + w_{i+\kappa}.$$

The factor swap supplies stochasticity and the eigenpair; saturation will ensure that the modified row still uses only one polygon side.

Lemma 2.4 (Positive interchange). *If $\beta_i = t > 0$ and $\beta_{i+1} = 0$, replacing v_i by c_i gives another saturated invariant N -gon with the same cyclic shift. Its coefficients satisfy*

$$\alpha'_i = 1, \quad \alpha'_{i+\kappa} = \alpha_i \alpha_{i+\kappa}, \quad \alpha'_j = \alpha_j \quad (j \notin \{i, i+\kappa\}). \quad (2.5)$$

The corresponding stochastic matrices are similar.

Proof. Write $\hat{B}$ for B with its i th row changed to e_i^T , and set $S = S_i(t)$, $H = C^\kappa \hat{B}$. Since the only other row that could use column i is row $i+1$, whose coefficient there is zero,

$$B = \hat{B}S, \quad A = HS, \quad A' = SH = SAS^{-1}.$$

Here $S^{-1} = (I - tE_i)/(1 - t)$, and both H and S are stochastic. Hence $v' = Sv$ satisfies $A'v' = \zeta v'$ and its convex hull is invariant. It has at most N vertices, hence exactly N by minimal order. Its displayed points remain in the original cyclic order: only v_i moves into its preceding side. Saturation applies.

Put $j = i + \kappa$, so $j \neq i$. Moving the elementary factor through C^κ gives $A' = C^\kappa S_j(t) \hat{B}$. Only row j of $\hat{B}$ is changed, to

$$t\hat{\beta}_{j-1}e_{j-2}^T + (t\hat{\alpha}_{j-1} + (1-t)\beta_j)e_{j-1}^T + (1-t)\alpha_j e_j^T.$$

The coefficients at $j-1$ and j are positive because $0 < t < 1$ and every $\hat{\alpha}$ is positive. This row represents a boundary contact of the new polygon. The face rule forces all its positive support into one side, and the vertices v'_{j-1}, v'_j already determine that side. It follows that $\hat{\beta}_{j-1} = 0$: otherwise the third vertex v'_{j-2} would belong to it. These three vertices are distinct because $N \geq 3$. This argument also covers $j-1 = i$, when the hatted coefficient is already zero.

The coefficient at j is $(1-t)\alpha_j = \alpha_i \alpha_j$, while all other rows agree with $\hat{B}$. This proves (2.5). The coefficient at the terminal vertex of each side remains positive, so the same shift and half-open assignment are retained. $\square$

The support $I = \{i : w_i > 0\} = \{i : \beta_i > 0\}$ therefore changes to $(I \setminus \{i\}) \cup \{i+\kappa\}$. Minimum branching will forbid a collision of two positive weights; the area choice will place their support in one interval.

Proposition 2.5 (Branch-minimal normal form). *The chosen polygon has, after cyclic relabelling,*

$$I = \{1, \dots, \varphi\}, \quad \varphi \geq \delta := \gcd(N, \kappa).$$

If $\varphi < N$, it is the deficit of a record of the rotation by κ : for some $h > 0$,

$$[h\kappa]_N = N - \varphi, \quad [t\kappa]_N < N - \varphi \quad (0 \leq t < h). \quad (2.6)$$

For $\varphi = N$, use the time-zero record.

Proof. Every κ -orbit meets I ; otherwise all contacts along that orbit would be endpoints, and their identities would give $v = \zeta^{N/\delta}v$. Suppose I is proper. At the last index i of any cyclic component, $\beta_i > 0$ and $\beta_{i+1} = 0$, so Lemma 2.4 applies. Moving the extreme vertex v_i into its preceding side gives a proper subpolygon. The support formula shows that its branching count cannot increase, and global minimality prevents it from decreasing.

Apply this operation separately at each component end of the chosen polygon P . The replacement point $c_i = \zeta v_{i-\kappa}$ has radius at most $\rho < 1$. If any radius-one vertex other than v_i remained, the new polygon would still be normalized and branch-minimal, with strictly smaller area. Therefore v_i must be the unique radius-one vertex. Two different component ends cannot both have this property. Hence I is one interval; since it meets every residue class modulo δ , its size is at least δ .

For the record assertion assume $\varphi < N$ and temporarily label the interval $F \cup \{0\}$, where $F = \{N - \varphi + 1, \dots, N - 1\}$. Let $a_t = [t\kappa]_N$, and let h be its first positive entrance into $\{N - \varphi, \dots, N - 1\}$. Such an entrance exists because this set of $\varphi \geq \delta$ consecutive residues contains a multiple of δ , and the orbit of zero consists of all those multiples.

Before that entrance, follow the branching contact initially at zero. In the state $F \cup \{a_t\}$ one has $a_t < N - \varphi$, so $a_t + 1 \notin F \cup \{a_t\}$ and the interchange is allowed at a_t . It moves this contact to a_{t+1} and leaves F in place. If $a_{t+1} \in F$, the two branching contacts coalesce and the branching count drops by one, contrary to its global minimum. Otherwise the count is unchanged, and the new state is $F \cup \{a_{t+1}\}$. These later polygons need not minimize area; minimum branching is preserved, since a rescaling would put any one of them back in the normalized class without changing its count.

At the first entrance, every target except $N - \varphi$ lies in F . The forbidden collision therefore gives $a_h = N - \varphi$, with all earlier residues smaller. Relabelling the original interval back to $\{1, \dots, \varphi\}$ leaves these residue displacements unchanged. $\square$

3 Return geometry

3.1 Arithmetic suspension

An endpoint contact gives the exact identity $\zeta v_i = v_{i+\kappa}$. Starting from an interior-contact index, these identities can be followed until the next such index is reached. We call the intervening list of indices a tower. Its height records how many applications of ζ have been contracted, while the unreduced index advance records how many turns have been made. Both quantities will enter the product equation. We first determine the towers by arithmetic alone.

The record basis reformulates Karpelevič's analysis of Euclidean remainders [3, §3, Lemma 2 and Theorems III–IV]. Slater studies related gap and return phenomena for rotations [29]; Hardy and Wright give the Farey background [16, Chapter III]. The finite suspension and its endpoint conventions are proved explicitly here.

Fix $1 \leq \kappa < N$, $\delta = \gcd(N, \kappa)$, and define $L(a, b) = a\kappa - bN$. A record time h is a strict record for $[h\kappa]_N$, with deficit $\nu = N - [h\kappa]_N$. At time zero take $V = (0, 1)$ and $\nu = N$; otherwise take $V = (h, [h\kappa/N])$. The last record has deficit δ .

Lemma 3.1 (Record basis). *For consecutive records $V = (h, b)$ and $V' = (q, p)$, whose deficits are $\nu > \nu'$, put $U = V' - V = (q, p)$ and $\Delta = \nu - \nu'$. Then*

$$\det(U, V) = 1, \quad L(U) = \Delta, \quad L(V) = -\nu, \quad q\nu + h\Delta = N, \quad \gcd(\Delta, \nu) = \delta. \quad (3.1)$$

In particular $q > 0$ and $0 < \Delta < \nu$.

Proof. The attainable residues are the multiples of δ , so the last deficit is δ . Every record vector is primitive: if $V = gW$, $g > 1$, then the earlier time of W would have the smaller positive deficit ν/g . Also

$$-\det(V, V') = \frac{h'\nu - h\nu'}{N} > 0.$$

If this integer is $D > 1$, complete the primitive vector V to an integer basis (V, W_0) by Bézout, choosing its orientation so that $V' = aV + DW_0$. Primitivity of V' gives $\gcd(a, D) = 1$. Thus

$$W = (\lceil a/D \rceil - a/D)V + D^{-1}V' = \lceil a/D \rceil V + W_0$$

is a nonzero integer point $\alpha V + \beta V'$ with $0 < \alpha, \beta < 1$. Replacing W by $V + V' - W$ if necessary gives $\alpha + \beta \leq 1$. Its time is strictly between zero and h' , while

$$0 < -L(W) = \alpha\nu + \beta\nu' < \nu.$$

It would therefore improve the first record before the second one, a contradiction. Thus $\det(U, V) = 1$. The remaining identities follow by applying L to the basis and expanding its determinant. In particular $L(\mathbb{Z}^2) = \delta\mathbb{Z} = \Delta\mathbb{Z} + \nu\mathbb{Z}$. $\square$

The identity $q\nu + h\Delta = N$ suggests $\nu - \Delta$ towers of height q and Δ towers of height $q + h$: their heights sum to N . The congruences determine where their tops return. To prove that these lists really partition all indices, it remains to exclude repetitions. The following argument does this using the cycles of the base permutation. It requires only the displayed integer relations, so it can also be used later with a different choice of bases.

Lemma 3.2 (Arithmetic suspension). *Suppose positive integers ν, Δ, q and $h \geq 0$ satisfy $1 \leq \Delta \leq \nu$, $\nu \leq N$,*

$$\begin{aligned} q\kappa &\equiv \Delta \pmod{N}, & h\kappa &\equiv -\nu \pmod{N}, \\ q\nu + h\Delta &= N, & \gcd(\Delta, \nu) &= \gcd(\kappa, N) = \delta. \end{aligned}$$

On the bases $J = \{1, \dots, \nu\}$ define

$$H_j = q + h\mathbf{1}_{\{j > \nu - \Delta\}}, \quad \sigma(j) = j + \Delta \pmod{\nu}$$

with values in J . Then

$$\pi(t, j) = [j + t\kappa]_N, \quad j \in J, \quad 0 \leq t < H_j,$$

is a bijection onto $\mathbb{Z}/N\mathbb{Z}$. The first return of addition by κ to J has time H_j and destination $\sigma(j)$.

Proof. Regard the pairs (t, j) as abstract states before identifying them with residues. Advance from (t, j) to $(t + 1, j)$ inside a tower, and from $(H_j - 1, j)$ to $(0, \sigma(j))$ at its top. Under π , an internal step is addition by κ . A top step has the same property: for a short tower $q\kappa \equiv \Delta$, and for a long tower $(q + h)\kappa \equiv \Delta - \nu$, exactly accounting for the wrap in $\sigma(j)$.

The base permutation has δ cycles, one for each residue class modulo δ . Each has ν/δ bases and Δ/δ long bases, since the long block has length Δ and both lengths are divisible by δ . Each resulting suspension cycle therefore has

$$q\nu/\delta + h\Delta/\delta = N/\delta$$

states. Addition by κ also has period N/δ , so the images of those states are one entire κ -orbit, without repetition. Different base cycles map to different residue classes modulo δ . This proves bijectivity. In particular, no internal state can represent an element of J , since that residue is already represented by its base state $(0, j)$. The displayed heights are therefore the first return times, including when $\nu = \Delta = 1$. $\square$

Apply the record lemma at the deficit φ supplied by Proposition 2.5, provided $\varphi > \delta$. The time-zero record has next record $(1, 1)$, so it is included. We obtain

$$q\kappa - pN = \Delta, \quad h\kappa - bN = -\varphi, \quad q\varphi + h\Delta = N, \quad 1 \leq \Delta < \varphi, \quad \gcd(\Delta, \varphi) = \delta. \quad (3.2)$$

Every internal destination in this suspension lies outside $I = \{1, \dots, \varphi\}$, so its contact equation is an endpoint identity. At the top, the destination belongs to I : the image is the contact on the side ending at $v_{\sigma(j)}$, rather than that vertex itself. Thus the arithmetic first returns give the geometric identities

$$\zeta^t v_j = v_{\pi(t,j)} \quad (0 \leq t < H_j), \quad \zeta^{H_j} v_j = c_{\sigma(j)}, \quad \zeta^h v_\varphi = v_0, \quad \zeta^q v_0 = c_\Delta. \quad (3.3)$$

Here the closing state is $\pi(h, \varphi) = 0$, with $h < H_\varphi = q + h$.

For example, take $N = 8$, $\kappa = 3$, and the record deficit $\varphi = 2$. The record occurs at $h = 2$, and the next record gives $q = 3$ and $\Delta = 1$. The two towers are

base j	H_j	indices before return	$\sigma(j)$
1	3	1, 4, 7	2
2	5	2, 5, 0, 3, 6	1

Each residue occurs once. For a contact representation with these data, the corresponding closing equations are $\zeta^3 v_1 = c_2$ and $\zeta^5 v_2 = c_1$; the second tower also contains $\zeta^2 v_2 = v_0$. We continue this example after Theorem 4.1, where the paths give both a polynomial and its real phase. The arithmetic construction alone does not assert the existence of an invariant polygon with prescribed contact data.

Reducing all indices modulo N would lose the winding needed to select the boundary branch. We therefore retain the lifted vertex arguments $\Phi_{i+N} = \Phi_i + 2\pi$ from (2.4). Along an endpoint path of length t , starting at index a ,

$$\Phi_a + t\theta_\zeta = \Phi_{a+t\kappa}. \quad (3.4)$$

If only its last contact is interior, the right-hand side is replaced by the open interval $(\Phi_{a+t\kappa-1}, \Phi_{a+t\kappa})$. The index $a + t\kappa$ is an integer, not merely a residue. Composition adds path lengths and index advances. Consequently contracting deterministic paths preserves the integer winding as well as the complex eigenvalue equation.

3.2 Face persistence and projective transfer

Karpelevič's proof uses a projective deformation to force adjacent return [3]. We retain that geometric purpose, using face persistence to supply its exposing lines. The projective facts needed below are derived in coordinates; Coxeter [12] gives the standard background on perspectivities and projectivities.

For an injective affine map with $F(P) \subseteq P$, let $G_P(x)$ be the minimal face of P containing x . Then

$$F(G_P(x)) \subseteq G_P(Fx), \quad \dim G_P(x) \leq \dim G_P(Fx).$$

Indeed, $x \in \text{relint } G_P(x)$, so every supporting functional at Fx pulls back to a functional constant on $G_P(x)$. This proves the inclusion, and injectivity gives the dimension inequality. In particular, a nonvertex can never become a vertex. In the plane this says that an image can pass from a side into the interior, but cannot pass from a side interior to a vertex. The half-open contact assignment gives a more precise conclusion when an entire side remains on the boundary: it identifies the endpoint that must remain a vertex.

Lemma 3.3 (Boundary persistence and one-sided exposure). *Assume the half-open contact assignment (2.3) for the invertible map F , and write $E_i = [v_{i-1}, v_i]$. If $r \geq 1$ is an integer and $F^r E_i \subseteq \partial P$, then*

$$F^t v_{i-1} = v_{i-1+t\kappa} \quad (0 \leq t \leq r).$$

Consequently, if $F^r v_i$ is in a side interior and v_{i-1} has a nonvertex image by time r , the inverse image of that side's line exposes v_i .

Proof. Invertibility and $F(P) \subseteq P$ imply $F(\text{int } P) \subseteq \text{int } P$. Thus no point of FE_i can be interior: its subsequent image under F^{r-1} is on the boundary. The nondegenerate boundary segment FE_i must lie in one side. Its endpoints, however, have the adjacent half-open assignments

$$Fv_{i-1} \in (v_{i+\kappa-2}, v_{i+\kappa-1}], \quad Fv_i \in (v_{i+\kappa-1}, v_{i+\kappa}].$$

The second endpoint is beyond their common vertex, so the first endpoint must equal that vertex. Consequently

$$Fv_{i-1} = v_{i+\kappa-1}, \quad \text{relint}(FE_i) \subseteq \text{relint } E_{i+\kappa}.$$

To repeat this argument, we need the whole next side, not just the subsegment FE_i . Choose $y \in \text{relint}(FE_i)$. It is a strict convex combination of the endpoints of $E_{i+\kappa}$, while $F^{r-1}y$ lies on the boundary. A supporting functional there is maximal on both images of those endpoints, because it is maximal on their strict convex combination. Hence $F^{r-1}E_{i+\kappa} \subseteq \partial P$. Induction on r now proves the asserted endpoint identities.

The pulled-back line supports P at v_i . If it also contained v_{i+1} , the first assertion applied to E_{i+1} would make $F^r v_i$ a vertex. If it contained v_{i-1} , the same assertion applied to E_i would make $F^r v_{i-1}$ a vertex, contrary to persistence of nonvertices. It therefore exposes v_i . $\square$

We next isolate the geometric deformation that will be applied to the exposing lines supplied by this lemma. Exposure provides the lines along which vertices may move; we still need a choice of motion that opens a contact inward. Its construction is local near the original polygon, but the sign of the deformation is determined by evaluating a projectivity at one remote point. Completed lines allow that evaluation even if an intermediate projection goes to infinity in the original affine chart.

These projective operations can be read as linear algebra in homogeneous coordinates: an affine point (x, y) represents the one-dimensional subspace spanned by $(x, y, 1)$ in $\mathbb{R}^3$, and a completed line is represented by a two-dimensional subspace. Projection from a centre c onto the line $\eta^T X = 0$ is induced by

$$X \mapsto \left(I - \frac{c\eta^T}{\eta^T c} \right) X.$$

If the centre is on neither line, this is an isomorphism between their two-dimensional subspaces, hence a fractional-linear map of line coordinates. An invertible 3×3 matrix changes the projective chart. Where its affine denominator has constant sign, it preserves segments and their interiors.

is an invertible projective change of chart, finite throughout P . Moreover,

$$\Phi((1-s)x + sy) = \frac{(1-s)\eta(x)\Phi(x) + s\eta(y)\Phi(y)}{(1-s)\eta(x) + s\eta(y)} \quad (0 \leq s \leq 1).$$

The positive weights show directly that convexity, segments, and side interiors are preserved.

The first coordinate has unique minimum zero at X_0 and unique maximum one at $X_{\ell+1}$. Each vertical slice of the polygon is an interval, so its two boundary chains between these endpoints are graphs. There are no vertical sides: a vertical supporting line would give a horizontal extremum, and both extrema are single vertices. Reflect the second coordinate if necessary to make the selected chain the lower graph. The internal exposing lines are nonvertical for the same reason. Consecutive genuine vertices give strictly increasing side slopes, and exposure puts the slope of L_i strictly between those of its two incident sides. Thus

$$X_i = (t_i, f_i), \quad t_0 < \cdots < t_{\ell+1}, \quad d_1 < \cdots < d_{\ell+1}, \quad d_i < s_i < d_{i+1},$$

where d_i is the slope of its i th side and s_i that of L_i . Set $s_1 = d_1$ and $h_i = t_i - t_{i-1} > 0$. Write the contacts and seek the successive remote projections in the form

$$C_i = X_i - \gamma_i h_i(1, d_i), \quad 0 < \gamma_i < 1, \quad Z_i = X_i - r_i(1, s_i).$$

The starting point $Z_1 = X_0$ gives $r_1 = h_1$. At each step the required collinearity is that of the vectors

$$\begin{aligned} C_i - Z_{i-1} &= (1 - \gamma_i)h_i(1, d_i) + r_{i-1}(1, s_{i-1}), \\ Z_i - C_i &= \gamma_i h_i(1, d_i) - r_i(1, s_i). \end{aligned}$$

Their determinant is zero. In reciprocal coordinates $x_i = 1/r_i$, expanding this determinant gives the affine transfer rule

$$x_i = \frac{(1 - \gamma_i)(s_i - d_i)}{\gamma_i(d_i - s_{i-1})}x_{i-1} + \frac{s_i - s_{i-1}}{\gamma_i h_i(d_i - s_{i-1})}. \quad (3.6)$$

Both its slope and its intercept are strictly positive, since $s_{i-1} < d_i < s_i$. Starting with $x_1 = 1/h_1 > 0$, it therefore gives finite positive x_i . The points defined by $r_i = 1/x_i$ satisfy the collinearity equations, so they are the unique projective intersections. This proves their finiteness without assuming it in advance. The positive intercept also gives the strict bounds

$$0 < r_i < \gamma_i h_i \frac{d_i - s_{i-1}}{s_i - s_{i-1}} < \gamma_i h_i \quad (2 \leq i \leq \ell).$$

For the final projection put $r = r_\ell$, $g = \gamma_\ell h_\ell$, $H = h_{\ell+1}$, $d = d_\ell$, $s = s_\ell$, and $D = d_{\ell+1}$. The signed line functional $G(W) = \det(X_{\ell+1} - Z_\ell, W - Z_\ell)$ satisfies

$$\begin{aligned} G(C_\ell) &= H(g - r)(D - s) + g(H + r)(s - d) > 0, \\ G(C_{\ell+1}) &= -\gamma_{\ell+1} H r (D - s) < 0. \end{aligned}$$

The opposite signs place the final intersection strictly between the two contacts, proving (3.5).

Return now to the original affine chart, where the deformation in the statement is defined. Use $z(t) = (1 - t)C_{\ell+1} + tC_\ell$ as affine coordinate on K . The map $\Pi \circ S^{-1}$ becomes

$$u(t) = \frac{at}{1 + bt}, \quad a \neq 0, \quad u(0) = 0, \quad 0 < u(1) < 1.$$

Consequently

$$t - u(t) = \frac{(1 - a)t + bt^2}{1 + bt}. \quad (3.7)$$

If $a \neq 1$, choose the sign of t so that $(1 - a)t > 0$. If $a = 1$, then $0 < u(1) < 1$ forces $b > 0$, and either sufficiently small nonzero sign works. The final moving line intersects K at $z(u(t))$. At zero, increasing the coordinate toward C_ℓ enters its inward half-plane: C_ℓ is strictly inward from the final side. More explicitly, choose a continuous signed affine functional for the moving line, positive on its inward side. Its value at $z(t)$ is $\Gamma(t)(t - u(t))$, with $\Gamma(0) > 0$. The positive factor persists near zero, so (3.7) proves both the inward motion and its first- or second-order opening. The calculation at the remote parameter 1 determines the sign of this local motion; it does not require the affine construction to remain finite throughout $[0, 1]$. $\square$

The local sign is now settled, including the case of a vanishing first variation. To use it in an invariant polygon, we must also propagate the motion along all vertex orbits and preserve every other terminal contact. The next proof separates these global constraints by their side indices.

Theorem 3.5 (No skipping in a saturated contact network). *Assume hereditary saturation as in Theorem 2.2, $N \geq 4$, and the return data (3.2)–(3.3) for the exact interval of interior contacts $B = \{1, \dots, \varphi\}$, with $\varphi > \gcd(N, \kappa)$. Then $\Delta = 1$.*

Proof. Write F for multiplication by the chosen multiplier, and recall

$$\sigma(j) = 1 + [j + \Delta - 1]_\varphi, \quad H_j = q + h\mathbf{1}_{j > \varphi - \Delta}, \quad F^{H_j}v_j = c_{\sigma(j)}, \quad F^h v_\varphi = v_0.$$

The first nonvertex image of base v_j occurs at time H_j ; all earlier tower steps are vertex identities. For $j \geq 2$, its predecessor v_{j-1} has already acquired a nonvertex image by that time, since $H_{j-1} \leq H_j$. Lemma 3.3 therefore gives

$$\mathcal{L}_j := F^{-H_j} \text{ aff } E_{\sigma(j)} \quad \text{exposes } v_j \quad (2 \leq j \leq \varphi).$$

This is also why the possible jump from height q to height $q + h$ causes no difficulty: a nonvertex image remains a nonvertex throughout the additional steps.

Suppose $\Delta > 1$. We shall deform the polygon while preserving invariance and making one image vertex interior, contradicting saturation. Choose the shorter of the two contact chains joining the seed to its target, with orientation recorded by ε :

$$(\varepsilon, \ell, X_i, a) = \begin{cases} (1, \Delta, v_i, q), & 2\Delta \leq \varphi + 1, \\ (-1, \varphi - \Delta + 1, v_{\varphi-i}, q + h), & 2\Delta > \varphi + 1, \end{cases} \quad 0 \leq i \leq \ell + 1.$$

Here $2 \leq \ell \leq (\varphi + 1)/2$. If $\varphi < N$, then $\ell + 1 \leq \varphi < N$; if $\varphi = N \geq 4$, then $\ell + 1 \leq (N + 3)/2 < N$. Thus the chain is proper. The endpoints are unmoved bases except possibly $X_0 = v_0$, which is fixed by the unmoved base v_φ through $v_0 = F^h v_\varphi$. Take $L_i = \mathcal{L}_j$ when $X_i = v_j$, $2 \leq i \leq \ell$; the indices lie in $\{2, \dots, \varphi\}$, so all these lines expose their vertices. Let C_i be the contact on $[X_{i-1}, X_i]$. In the two orientations,

$$(C_\ell, C_{\ell+1}) = (c_\Delta, c_{\Delta+1}) \quad \text{or} \quad (c_\Delta, c_{\Delta-1}),$$

and the tower identities give $F^a X_0 = C_\ell$, $F^a X_1 = C_{\ell+1}$.

To extend the chain motion to the polygon, we must control both the vertex identities inside towers and the side contacts at their ends. Only sides E_i with $i \in B$ carry terminal contacts. Their endpoints are bases, except $v_0 = F^h v_\varphi$, which stays fixed. Let b be the index of X_1 , M the indices of $X_1, \dots, X_\ell$, $k_* := \sigma(b)$ the final side index, and J the internal side indices. Reading each side by its ending vertex, in $\mathbb{Z}/\varphi\mathbb{Z}$ one has

$$\sigma(M) = \{k_* + \varepsilon t : 0 \leq t < \ell\}, \quad J = \{k_* - \varepsilon r : 1 \leq r < \ell\}. \quad (3.8)$$

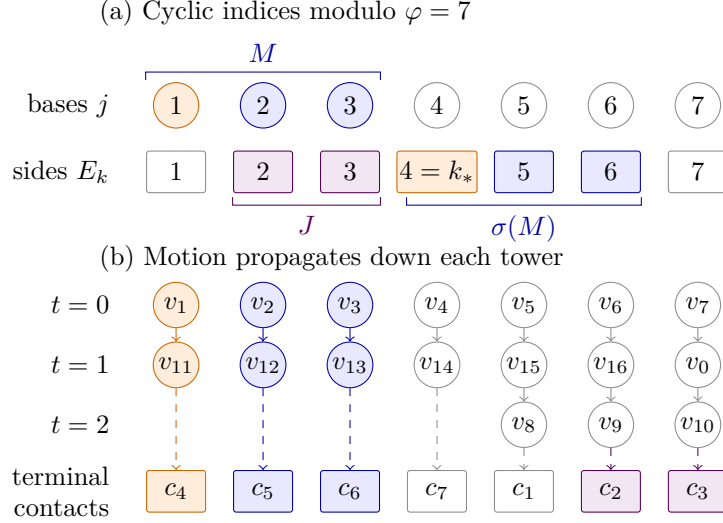


Figure 2: The arithmetic schema $(N, \kappa, \varphi, \Delta, q, h) = (17, 10, 7, 3, 2, 1)$, with $\ell = 3$ and $\sigma(j) = 1 + [j + 2]_7$. The moving bases are $M = \{1, 2, 3\}$; the internal chain sides $J = \{2, 3\}$ are disjoint from their targets $\sigma(M) = \{4, 5, 6\}$. Solid arrows are exact vertex images under F ; dashed arrows are the terminal side contacts before deformation. Coloured towers move by (3.9); uncoloured towers stay fixed. The two violet contacts have fixed sources and moving side lines, the two blue contacts have moving sources and fixed side lines, and only the orange pair has both source and target line moving. This is an index and incidence diagram, not a claimed invariant polygon with $\Delta > 1$.

These arcs are disjoint, since an intersection would require $\varphi \mid (t+r)$ with $1 \leq t+r \leq 2\ell-2 < \varphi$. Also $\sigma(b) = k_*$ and no other moving source targets that side. Thus a changing internal side has a fixed terminal source, while every moving source except the seed has a fixed target line. Figure 2 shows this separation together with the tower extension used below.

Apply Lemma 3.4 with $S = F^a|_{\Lambda_1}$. Denote its deformation parameter by τ , to distinguish it from the integer tower time t . Write $\hat{v}_j(\tau) = X_i(\tau)$ when $v_j = X_i$ for $1 \leq i \leq \ell$, and keep the other bases fixed. The tower bijection extends these choices to every vertex by

$$\hat{v}_{\pi(t,j)}(\tau) = F^t \hat{v}_j(\tau) \quad (0 \leq t < H_j). \quad (3.9)$$

In particular, each internal image is exactly the next vertex of its tower. The remaining images are $F^{H_j} \hat{v}_j(\tau)$, whose original target sides are $E_{\sigma(j)}$.

For an internal chain side, indexed by J , the terminal source is fixed: the disjointness in (3.8) says that its index is outside M . Its image is the fixed contact C_i , and the chain construction keeps that contact on $\text{aff}(X_{i-1}(\tau), X_i(\tau))$.

Next consider a moving source X_i with $i \geq 2$. Its target index belongs to $\sigma(M) \setminus \{k_*\}$, hence is neither an internal chain side nor the final side. That target line stays fixed. Indeed, among sides carrying terminal contacts, the only changing lines are the internal and final chain lines: the initial line stays Λ_1 , and the remaining endpoints are fixed bases or v_0 . Since $X_i(\tau)$ stays on the pulled-back line L_i , its terminal image stays on its fixed target line.

Every terminal side outside $J \cup \sigma(M)$ has a fixed source and a fixed line, so its incidence is unchanged. Finally, the seed is the unique moving source with target k_* ; its terminal image is $Y(\tau) = F^a X_1(\tau)$, and the final moving side is $[X_\ell(\tau), X_{\ell+1}]$. This is precisely the incidence that the holonomy lemma moves strictly inward.

All assertions so far concern exact identities or line incidences. We now restrict τ to a sufficiently small neighborhood of zero. The vertices remain distinct and strictly convex,

and each preserved terminal incidence stays in the relative interior of its side, since it starts there. The exceptional image $Y(\tau)$ starts in the relative interior of the final side, so it remains strictly inside all other side half-planes. Choose an arbitrarily small nonzero parameter given by the holonomy lemma to place it inward of the final side as well. Thus every image vertex belongs to the deformed N -gon, with exactly one strictly interior. Invertibility makes it a vertex of the image polygon. This contradicts hereditary saturation. $\square$

4 The product and its winding

The no-skipping theorem leaves only two return types. If $\varphi > \delta = \gcd(N, \kappa)$, there are more interior contacts than shift orbits; no skipping forces $\Delta = 1$, hence $\delta = 1$. Otherwise the contact interval has length δ . In the coprime case, including a single interior contact, the inverse shift gives a successor suspension. In the remaining case, each shift orbit has exactly one interior contact, and the return recurrence is reversed after conjugation.

In both cases the objective is the same: obtain equations linking successive base vertices, multiply them to eliminate those vertices, and telescope their lifted arguments. The last operation is needed because multiplying complex equations determines a phase only modulo 2π ; the boundary branch requires its actual integer winding.

The cyclic recurrence with side-dependent coefficients already appears in Karpelevič [3, Theorem VI and equation (13), pp. 375–376]. There the coefficients are equalized before the vertices are eliminated. Here they remain unequal through the geometric reduction and multiplication; their equality will be a conclusion of the convex comparison.

We use the elementary Farey criterion: reduced fractions $p/q < r/s$ with denominators at most N are consecutive in F_N exactly when $rq - ps = 1$ and $q + s > N$. Here is a proof including the denominator bound. Put $U = (q, p)$, $V = (s, r)$, and $D = \det(U, V) > 0$. Because U is primitive, Bézout's identity supplies an integer vector W with $\det(U, W) = 1$. Write $V = aU + DW$; primitivity of V gives $\gcd(a, D) = 1$. If $D > 1$, the integer point

$$Q = \left(\left\lceil \frac{a}{D} \right\rceil - \frac{a}{D} \right) U + \frac{1}{D} V = \left\lceil \frac{a}{D} \right\rceil U + W$$

has both coefficients strictly between zero and one. If their sum exceeds one, replace Q by $U + V - Q$. In either case $Q = \xi U + \eta V$ with $\xi, \eta > 0$ and $\xi + \eta \leq 1$. Writing $Q = (d, c)$, its slope c/d lies strictly between p/q and r/s , while $0 < d \leq \max(q, s) \leq N$. Reducing that fraction only decreases its denominator. Thus consecutive fractions must have $D = 1$. They must also have $q + s > N$, since otherwise their mediant is an intermediate fraction of allowable denominator.

Conversely, if $D = 1$, any intermediate fraction c/d satisfies

$$(d, c) = (rd - sc)(q, p) + (qc - pd)(s, r).$$

The two coefficients are positive integers, so $d \geq q + s$. When $q + s > N$, no fraction of F_N can lie between the pair.

Theorem 4.1 (Product reduction). *Assume (2.2) with $N \geq 4$. For one orientation $\omega \in \{\zeta, \bar{\zeta}\}$, write $\omega = \rho e^{i\vartheta}$ with $0 < \vartheta < 2\pi$. There are consecutive fractions*

$$\frac{p}{q} < \frac{\vartheta}{2\pi} < \frac{r}{s} \quad \text{in } F_N, \quad q < s,$$

and, with $m = \lfloor N/q \rfloor$ and $e = s - mq$, numbers $0 \leq \beta_j < 1$, $\alpha_j = 1 - \beta_j$, such that

$$\omega^e \prod_{j=1}^m (\omega^q - \beta_j) = \prod_{j=1}^m \alpha_j. \quad (4.1)$$

For $A = q\vartheta - 2\pi p$ and $M = \text{Arg}(\omega^q - 1)$, the same factors have arguments $u_j \in [A, M]$ satisfying

$$0 < A < M < \pi, \quad e\vartheta + \sum_{j=1}^m u_j = 2\pi(r - mp). \quad (4.2)$$

When $e < 0$, (4.1) equivalently means the polynomial identity obtained by multiplying it by ω^{mq} .

Proof. Take the branch-minimal contact representation. By the no-skipping theorem, $\varphi > \delta$ implies $\Delta = 1$, hence $\delta = 1$.

Coprime shift. Suppose $\delta = 1$. Let $q \in \{1, \dots, N-1\}$ be the inverse of κ modulo N and write $q\kappa - pN = 1$. When $\varphi > 1$, this is precisely the q in (3.2), since $q\varphi + h = N$ forces $q < N$. Thus $m = \lfloor N/q \rfloor \geq \varphi$; for $\varphi = 1$ the same inequality is automatic. Put $e = N - mq$ and $g = \kappa - mp$. Then

$$q\kappa - pN = 1, \quad e\kappa - gN = -m, \quad qm + e = N.$$

Lemma 3.2, now with $(\nu, \Delta, h) = (m, 1, e)$, gives a successor suspension whose bases contain all interior contacts. Its endpoint paths and top contacts yield

$$(\zeta^q - \beta_j)v_{j-1} = \alpha_j v_j \quad (1 \leq j \leq m), \quad \zeta^e v_m = v_0. \quad (4.3)$$

For $2 \leq j \leq m$, the q -step path from v_{j-1} ends at c_j . The remaining return from v_m has length $q + e$ and passes through v_0 after e steps. Splitting that return at v_0 gives the same q -step recurrence for $j = 1$ and the displayed closing identity. Thus the formulas also cover $m = 1$. The additional bases $j > \varphi$ merely subdivide deterministic paths: their coefficients are $\beta_j = 0$, so no new interior contact has been introduced.

The path from v_0 to c_1 has length q , endpoint intermediate destinations, and lifted advance $q\kappa = pN + 1$. Therefore

$$0 < q\theta_\zeta - 2\pi p < \Phi_1 - \Phi_0.$$

Together with (2.4), this places $\theta_\zeta/(2\pi)$ strictly between p/q and κ/N . Their determinant is one, their denominators are ordered $q < N$, and their sum exceeds N . Thus these are the required Farey neighbors, with $\omega = \zeta$, $r = \kappa$, $s = N$. Because $\alpha_j > 0$, the first recurrence in (4.3) identifies the argument of its factor with the angular gap $u_j = \Phi_j - \Phi_{j-1} \in (0, \pi)$. These choices telescope as real numbers. The endpoint closing path has lifted advance $m + e\kappa = gN$, so

$$\Phi_m + e\theta_\zeta = \Phi_0 + 2\pi g.$$

For $e = 0$ this is just the periodic extension of the vertex arguments. Multiplying the recurrences and telescoping these arguments proves both required identities.

One contact in each of several orbits. The only remaining case is $\varphi = \delta \geq 2$. Put $L = N/\delta$ and $K = \kappa/\delta$. Choose $1 \leq h < L$ with $Kh \equiv -1 \pmod{L}$, and set

$$b = \frac{Kh + 1}{L}, \quad s_0 = N - h, \quad r_0 = \kappa - b.$$

Each orbit has length L and just one interior contact. Thus

$$(\zeta^L - \alpha_i)v_i = \beta_i v_{i-1} \quad (1 \leq i \leq \delta), \quad \zeta^h v_\delta = v_0. \quad (4.4)$$

The closing path has no interior destination, because it remains in the residue-zero orbit and $h < L$. Its exact lift, and the lifts of the L -step returns, are

$$\Phi_\delta + h\theta_\zeta = \Phi_0 + 2\pi b, \quad 0 < 2\pi K - L\theta_\zeta < \gamma_i \quad (1 \leq i \leq \delta).$$

Writing $\Gamma = \sum_{i=1}^\delta \gamma_i$, these give

$$s_0\theta_\zeta - 2\pi r_0 = \Gamma - \delta(2\pi K - L\theta_\zeta) > 0, \quad \frac{r_0}{s_0} < \frac{\theta_\zeta}{2\pi} < \frac{K}{L}.$$

Here $1 \leq b \leq K$, $Ks_0 - r_0L = 1$, and $L + s_0 > N$, so the two endpoint fractions are reduced Farey neighbors of order N .

Conjugate (4.4) and reverse its list. Explicitly, set

$$\omega = \bar{\zeta}, \quad \vartheta = 2\pi - \theta_\zeta, \quad W_j = \bar{v}_{\delta-j}, \quad (p, q, r, s) = (L - K, L, s_0 - r_0, s_0).$$

Then $q < s$, since $\delta \geq 2$ and $h < L$, and $m = \delta$, $e = -h$. With $\tilde{\beta}_j = \alpha_{\delta-j+1}$, $\tilde{\alpha}_j = \beta_{\delta-j+1}$, the recurrences become

$$(\omega^q - \tilde{\beta}_j)W_{j-1} = \tilde{\alpha}_j W_j, \quad \omega^{-h}W_m = W_0.$$

Their factor arguments are $u_j = \gamma_{\delta-j+1}$; hence

$$e\vartheta + \sum_j u_j = -h(2\pi - \theta_\zeta) + 2\pi b - h\theta_\zeta = 2\pi(b - h) = 2\pi(r - mp).$$

Multiplication supplies the product, with the tildes then dropped.

Finally, in both cases Farey adjacency gives $0 < A < 2\pi/s < \pi$. The point $\omega^q - t$ stays in the upper half-plane for $0 \leq t \leq 1$, and its argument strictly increases from A to M . Since each coefficient is less than one, every previously selected argument lies in $[A, M)$. The polynomial and the real lift have therefore been obtained from exactly the same return paths. $\square$

The eight-state example, continued

Return to the towers with $N = 8$, $\kappa = 3$, and $\varphi = 2$ in Section 3.1. Here $q = 3$, $m = 2$, and $e = 2$, so the successor construction requires no additional bases. The five-step return splits at its intermediate state v_0 :

$$(\zeta^3 - \beta_1)v_0 = \alpha_1 v_1, \quad (\zeta^3 - \beta_2)v_1 = \alpha_2 v_2, \quad \zeta^2 v_2 = v_0.$$

Consequently the product and its lifted phase are

$$\zeta^2(\zeta^3 - \beta_1)(\zeta^3 - \beta_2) = \alpha_1 \alpha_2, \quad 2\theta_\zeta + u_1 + u_2 = 2\pi.$$

The real equality is visible without taking an argument of the product: $u_j = \Phi_j - \Phi_{j-1}$, and the closing path gives $\Phi_2 + 2\theta_\zeta = \Phi_8 = \Phi_0 + 2\pi$. The endpoint data are $(p, q, r, s) = (1, 3, 3, 8)$, so the Farey interval is $(1/3, 3/8)$ and $r - mp = 1$, exactly the winding just obtained.

To see the scalar calculation as well, take $\theta_\zeta = 5\pi/7$. Then $A = B = \pi/7$, and the scalar equation of the next section reduces to

$$\rho^4 + \rho^3 = 2 \cos(\pi/7).$$

Its left side increases from zero to two, so there is a unique $\rho \in (0, 1)$. Set

$$\beta_1 = \beta_2 = \beta = \frac{1}{1 + \rho}, \quad \alpha_1 = \alpha_2 = \alpha = \frac{\rho}{1 + \rho}.$$

For $\zeta = \rho e^{5\pi i/7}$, the scalar equation gives $\zeta^3 - \beta = \beta e^{2\pi i/7}$. Thus both factor arguments are $2\pi/7$, and both the product and its real phase hold exactly. For orientation only, $\rho \simeq 0.97061308$; this numerical value is not used in the proof.

There is also a direct eight-state realization. On states $0, \dots, 7$, give the paths

$$0 \longrightarrow 3 \longrightarrow 6, \quad 1 \longrightarrow 4 \longrightarrow 7, \quad 2 \longrightarrow 5 \longrightarrow 0$$

unit weights, and give the remaining rows the transitions

$$6 \xrightarrow{\beta} 0, \quad 6 \xrightarrow{\alpha} 1, \quad 7 \xrightarrow{\beta} 1, \quad 7 \xrightarrow{\alpha} 2.$$

Starting with $v_0 = 1$, the three displayed recurrences determine v_1, v_2 , and the unit paths determine the other five coordinates. They give a nonzero eigenvector of the resulting stochastic matrix at ζ . This proves feasibility at the stated radius. Its outermostness follows from the convex bound and Farey comparison below, which are independent of this example. Nor does this matrix construction alone assert that its displayed eigenvector coordinates are all polygon vertices; the earlier tower illustration remains conditional on the contact representation.

5 The scalar bound and its attainment

The product reduction supplies coefficients that may differ from side to side. Its real phase fixes the mean of the factor arguments, while its modulus fixes the sum of a convex function of those arguments. Their comparison will give the scalar radius and identify equality.

The one-factor logarithmic convexity was established for the restricted n -cycle family in [34, Lemmas 4.2 and 5.2]. Here normalizing each factor places it on one affine line; the radial distance to that line gives the convex function applied to the unequal product supplied by Theorem 4.1. The scalar equation that results is equivalent to the prescribed-ray description in [20, Theorems 1.2 and 4.3; Lemma 4.4]; our argument obtains it before identifying the stochastic boundary.

For consecutive fractions $f < g$ in $F_n \cap [0, 1/2]$, orient the coordinate by the smaller endpoint denominator. Thus either

$$(p/q, r/s, y) = (f, g, x), \quad \text{or} \quad (p/q, r/s, y) = (1 - g, 1 - f, 1 - x),$$

where $q < s$, $rq - ps = 1$, and $x = \theta/(2\pi) \in (f, g)$. In this section put

$$m = \lfloor n/q \rfloor, \quad A = 2\pi(qy - p), \quad B = \frac{2\pi(r - sy)}{m}. \quad (5.1)$$

The coordinate change conjugates the complex number and leaves its modulus unchanged. Writing $t = (y - p/q)/(r/s - p/q)$ gives

$$A = \frac{2\pi t}{s}, \quad B = \frac{2\pi(1 - t)}{mq}, \quad \frac{A + B}{2\pi} = \frac{t}{s} + \frac{1 - t}{mq}. \quad (5.2)$$

For $n \geq 3$, $s \geq 3$ and $mq \geq 2$; consequently $A, B > 0$ and $A + B < \pi$.

Definition 5.1. For an interior Farey ray, let $K_n(\theta)$ be the unique positive solution ρ of

$$\rho^{s/m} \sin A + \rho^q \sin B = \sin(A + B). \quad (5.3)$$

For $n \geq 4$, set $K_n(\theta) = 1$ on the Farey rays themselves. For order three we use the interior scalar roots without prescribing a continuous endpoint value on the negative real ray.

Existence and uniqueness follow because the left side increases strictly from zero, and at $\rho = 1$ exceeds the right side by $4 \sin(A/2) \sin(B/2) \sin((A+B)/2) > 0$. Thus $0 < \rho < 1$. Continuity on each open interval follows by passing to limits in the equation and using uniqueness. For $n \geq 4$, $mq \geq 3$; letting A or B tend to zero in (5.3) therefore gives the endpoint limit $\rho \rightarrow 1$. Hence K_n is positive and continuous on $[0, \pi]$.

Theorem 5.2 (Convex phase comparison). *Suppose $p/q < \vartheta/(2\pi) < r/s$, $rq - ps = 1$, $q < s$, and $m \geq 1$. Put*

$$A = q\vartheta - 2\pi p, \quad B = (2\pi r - s\vartheta)/m, \quad e = s - mq,$$

and assume $A + B < \pi$. Let $\omega = \rho e^{i\vartheta}$, $0 < \rho < 1$, and suppose $0 \leq \beta_j < 1$ satisfy both

$$\omega^e \prod_{j=1}^m (\omega^q - \beta_j) = \prod_{j=1}^m (1 - \beta_j), \quad e\vartheta + \sum_{j=1}^m u_j = 2\pi(r - mp), \quad (5.4)$$

where $u_j = \text{Arg}(\omega^q - \beta_j) \in [A, M)$ and $M = \text{Arg}(\omega^q - 1)$. Then

$$\rho^{s/m} \sin A + \rho^q \sin B \leq \sin(A + B). \quad (5.5)$$

Equality holds precisely when all β_j agree. In particular, ρ is bounded by the scalar root (5.3).

Proof. Write $w = \omega^q = \rho^q e^{iA} = x + iy$, so $y > 0$ and $x < 1$. Normalize the factors by their positive coefficients:

$$g_j = \frac{w - \beta_j}{1 - \beta_j}, \quad c = \frac{1 - x}{y} > 0, \quad \mathcal{D}(u) = \cos u + c \sin u.$$

Every g_j lies on the line $\Re g + c\Im g = 1$. Its argument is the same $u_j \in (0, \pi)$ as that of $w - \beta_j$, and hence $|g_j| \mathcal{D}(u_j) = 1$. The logarithm is defined on a common interval: on $(0, \pi)$, $\mathcal{D}(u)/\sin u = c + \cot u$ decreases strictly, so positivity at the largest u_j implies positivity throughout their convex hull.

Put $F = -\log \mathcal{D}$, so $F(u_j) = \log |g_j|$. The identity $\mathcal{D}'' = -\mathcal{D}$ and differentiation of the factor argument give

$$F''(u) = 1 + \left(\frac{\mathcal{D}'(u)}{\mathcal{D}(u)} \right)^2 \geq 1, \quad (5.6)$$

$$\frac{du}{d\beta} = \frac{\rho^q \sin A}{|\omega^q - \beta|^2} > 0.$$

The real phase fixes the mean argument, while the modulus of $\omega^e \prod_j g_j = 1$ fixes the sum of these logarithmic distances:

$$\bar{u} := \frac{1}{m} \sum_j u_j = A + B, \quad \sum_j F(u_j) = -e \log \rho.$$

Their mean lies in the same positive interval, so Jensen's inequality gives $F(\bar{u}) \leq -(e/m) \log \rho$, or $\mathcal{D}(A + B) \geq \rho^{e/m}$. Now

$$\mathcal{D}(A + B) = \frac{\sin(A + B) - \rho^q \sin B}{\rho^q \sin A}.$$

Multiplication by $\rho^q \sin A > 0$, with $q + e/m = s/m$, proves (5.5). Strict convexity gives equality exactly when all u_j agree; the strict increase of u in β makes this equivalent to equal coefficients. Strict increase of the scalar left side then proves the radius bound. $\square$

5.1 Realization by a cycle determinant

We use the block graph of Johnson and Paparella [5, Theorem 3.2] and the independent-weight cycle expansion already given in the supplied manuscript [1, Theorem 7.6]. Its two types of cycles determine the full characteristic polynomial. Specializing to equal weights proves attainment; Appendix B records the additional structure and endpoint behavior of the matrix family.

Let ρ solve (5.3), and define

$$\alpha = \frac{\rho^{s/m} \sin A}{\sin(A+B)}, \quad \beta = \frac{\rho^q \sin B}{\sin(A+B)}. \quad (5.7)$$

Then $\alpha, \beta \in (0, 1)$ and $\alpha + \beta = 1$. For $\omega = \rho e^{2\pi i y}$, direct real and imaginary parts give

$$\omega^q - \beta = \frac{\rho^q \sin A}{\sin(A+B)} e^{i(A+B)} = \alpha \rho^{q-s/m} e^{i(A+B)}. \quad (5.8)$$

As $m(A+B) = (mq-s)2\pi y + 2\pi(r-mp)$, it follows that

$$\omega^s (\omega^q - \beta)^m = \alpha^m \omega^{mq}. \quad (5.9)$$

Thus the scalar curve lies on the selected Ito branch, with its phase fixed as a real identity.

For $n \geq 4$, the parameter β traverses this scalar arc exactly once. We verify this in the oriented argument $\vartheta = 2\pi y$, increasing from $2\pi p/q$ to $2\pi r/s$. Let $\mathcal{R}(\rho, \vartheta)$ be the difference between the two sides of (5.3). On the open arc,

$$\mathcal{R}_\rho = \frac{s}{m} \rho^{s/m-1} \sin A + q \rho^{q-1} \sin B > 0.$$

The already established continuity of ρ , followed by the one-variable mean-value theorem applied to the scalar equations at ϑ and $\vartheta + h$, gives $\rho' = -\mathcal{R}_\vartheta / \mathcal{R}_\rho$. This identity first makes ρ continuously differentiable; repeated differentiation makes it smooth. Formula (5.7) therefore makes $\beta(\vartheta)$ smooth as well.

Set $e = s - mq$ and

$$H(\omega, \beta) = \omega^e (\omega^q - \beta)^m - (1 - \beta)^m.$$

Although e may be negative, this is holomorphic in $\omega \neq 0$. Differentiation gives

$$H_\omega = \omega^{e-1} (\omega^q - \beta)^{m-1} (s\omega^q - e\beta).$$

Every factor is nonzero on the selected open arc: $\text{Im}(\omega^q) = \rho^q \sin A > 0$, and $e\beta$ is real. Thus the selected root is simple, also in the Ito polynomial $\omega^{mq} H$, since $\omega \neq 0$. Differentiating $H(\omega(\vartheta), \beta(\vartheta)) = 0$ gives

$$H_\omega \omega' + H_\beta \beta' = 0, \quad \omega' = e^{i\vartheta} (\rho' + i\rho) \neq 0.$$

If β' vanished, this identity would contradict $H_\omega \neq 0$. Consequently β' has a constant nonzero sign. Since $s, mq \geq 3$, the endpoint limits $\rho \rightarrow 1$ and (5.7) give $\beta \rightarrow 1$ at the left endpoint and $\beta \rightarrow 0$ at the right. Hence β decreases strictly from one to zero and gives a one-to-one parametrization of the whole arc. The monotonicity refers to the oriented coordinate ϑ ; conjugation reverses the direction along the original upper-half-plane arc when required.

Theorem 5.3 (Independent-parameter cycle realization). *Let m, q, s be positive integers with $s > (m - 1)q$, and put $n_0 = \max\{mq, s\}$. For arbitrary*

$$0 \leq \beta_j \leq 1, \quad \alpha_j = 1 - \beta_j \quad (j \in \mathbb{Z}/m\mathbb{Z}),$$

there is a row-stochastic matrix $A(\beta)$ of order n_0 , whose entries depend affinely, hence continuously, on the parameters, with

$$\det(tI - A(\beta)) = t^{n_0 - mq} \prod_{j=0}^{m-1} (t^q - \beta_j) - t^{n_0 - s} \prod_{j=0}^{m-1} \alpha_j. \quad (5.10)$$

Proof. We build m local cycles of length q , joined by a connection cycle of length s . Skipping vertices at one entry, or inserting vertices into one connection, adjusts the second length. Create m blocks $v_{j,0}, \dots, v_{j,q-1}$, indexed cyclically, with deterministic transitions $v_{j,k} \rightarrow v_{j,k+1}$. At each terminal vertex place a return edge of weight β_j to $v_{j,0}$ and a connection of weight α_j to the next block, adding weights if edges coincide. If $s \leq mq$, let the connection entering block zero end at $v_{0,mq-s}$ and all other connections enter the initial vertices; the hypothesis gives $0 \leq mq - s < q$, so this entry belongs to block zero. The graph has mq vertices in this case. If $s > mq$, let all connections enter initial vertices and insert $s - mq$ new vertices into the connection entering block zero, retaining its weight α_{m-1} on the first edge and weight one on the rest. There are now s vertices. Every row is deterministic or has outgoing weights α_j, β_j summing to one. The matrix is therefore stochastic of order n_0 , with affine dependence on the parameters.

Figure 3 shows both modifications with $m = 2$, $q = 3$, and common weights $\beta_j = \beta$, $\alpha_j = \alpha$. There $\delta_0 = mq - s$ denotes the entry offset, or minus the number of inserted vertices when it is negative.

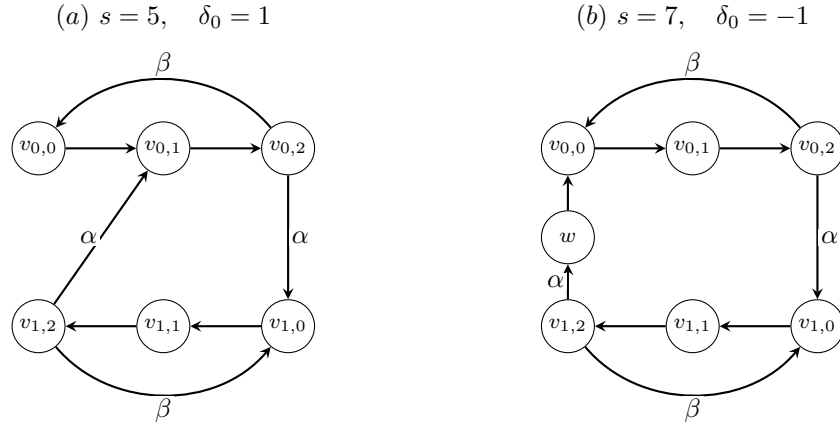


Figure 3: Two cyclic-transfer realizations with $m = 2$ and $q = 3$. Unlabelled arrows have weight one and $\alpha + \beta = 1$. In (a) the connection into block zero skips $v_{0,0}$, giving a connection cycle of length five on six vertices. In (b) the added vertex w gives a connection cycle of length seven on seven vertices. Both graphs retain the two three-cycles closed by the return edges.

If $m = 1$ and $s = q$, the two terminal edges coincide and have total weight one. The matrix is the deterministic q -cycle, with polynomial $t^q - 1 = t^q - \beta_0 - \alpha_0$, as asserted. In all other cases the two terminal destinations are distinct.

Every simple directed cycle contains a terminal. If it chooses a return there, the deterministic block path takes it back to the same terminal, so it is precisely that block's q -cycle. If it chooses no return, it follows the connections through every block and forms the single s -cycle. Thus the complete list consists of the m disjoint block cycles, of weights β_j , and

the connection cycle, of weight $\prod_j \alpha_j$. The latter meets every block cycle at its terminal. This description includes $q = 1$, when block cycles are loops, and $m = 1, s < q$, when the connection closes a shorter cycle within its block.

In the Leibniz expansion of $\det(tI - A)$, a collection of k vertex-disjoint directed cycles covering v vertices contributes its weight times $(-1)^k t^{n_0 - v}$. Indeed, a cycle of length d has permutation sign $(-1)^{d-1}$ and contributes d negative entries. At a fixed point, expanding $t - a_{ii}$ selects either an unused vertex or a loop, so the same rule applies to cycles of length one. The collections of block cycles therefore contribute

$$\sum_{J \subseteq \{0, \dots, m-1\}} (-1)^{|J|} t^{n_0 - q|J|} \prod_{j \in J} \beta_j = t^{n_0 - mq} \prod_{j=0}^{m-1} (t^q - \beta_j).$$

The connection cycle can occur only alone and contributes $-t^{n_0 - s} \prod_j \alpha_j$. Their sum is (5.10). Zero edge weights simply make the corresponding contributions vanish, so the argument includes every endpoint tuple. Being a polynomial identity, it includes zero roots and their algebraic multiplicities as well. $\square$

In particular, take all $\beta_j = \beta \in (0, 1)$ and $\alpha = 1 - \beta$. Every nonzero root λ of

$$\lambda^s (\lambda^q - \beta)^m = \alpha^m \lambda^{mq} \quad (5.11)$$

is a zero of (5.10), hence an eigenvalue of this same stochastic matrix of order $\max\{mq, s\}$.

For consecutive order- n Farey denominators, $q + s > n \geq mq$, so the theorem applies and its order is at most n . Choose (5.7) and use (5.9); conjugation handles the reflected coordinate and an identity block gives order n . The endpoint roots are realized by q - and s -cycles. Consequently $K_n(\theta)e^{i\theta} \in \Theta_n$ on every ray for $n \geq 4$.

Combining Theorem 4.1 with Theorems 5.2 and 5.3 already identifies every radial maximum at its first order of occurrence $N \geq 4$ with K_N . At such a maximum, the unequal parameters supplied by the geometric reduction must all agree. The remaining issue is to compare different orders without assuming the desired boundary theorem.

6 Farey refinement and completion

We need only two comparison operations: increase the factor count, or insert a mediant. The first is an immediate consequence of strict convexity.

Lemma 6.1 (Adding a factor). *Fix $p/q < y < r/s$, $rq - ps = 1$, $q < s$. For an integer $m \geq 1$, suppose $A = 2\pi(qy - p) > 0$, $B_m = 2\pi(r - sy)/m > 0$, and $A + B_m < \pi$. The scalar root for count $m + 1$ is strictly larger than the one for count m .*

Proof. For the old root, (5.8) gives an equal-parameter product with all m arguments equal to $A + B_m$. Append $\beta_{m+1} = 0$, whose factor is ω^q and whose argument is A . The product remains valid after changing the exterior exponent from $s - mq$ to $s - (m + 1)q$, and its real phase becomes

$$[s - (m + 1)q]2\pi y + m(A + B_m) + A = 2\pi[r - (m + 1)p].$$

The enlarged tuple is unequal, since the old parameter lies in $(0, 1)$. Theorem 5.2, now with $m + 1$ factors, gives strict scalar inequality at the old root. The new scalar left side is strictly increasing, so its root is larger. $\square$

The mediant comparison will reduce to the sign of a polynomial divided difference. The following elementary segment identity supplies that sign. Its imaginary part has a geometric meaning: for complex vectors v, w , $\text{Im}(\bar{v}w)$ is their planar determinant, so its sign locates w relative to the oriented line in direction v .

Lemma 6.2 (A segment comparison). *Let f be holomorphic near the segment $[a, b] \subset \mathbb{C}$, with $a \neq b$, and let $\chi \in \mathbb{R}$. If $\operatorname{Im}(e^{i\chi} f'(z)) > 0$ on $[a, b]$, then*

$$\operatorname{Im}\left(e^{i\chi}(\bar{a} - \bar{b})(f(a) - f(b))\right) > 0.$$

Proof. Parametrize the segment by $z(t) = b + t(a - b)$, $0 \leq t \leq 1$. Apply the ordinary one-variable fundamental theorem of calculus to the real and imaginary parts of $f(z(t))$; its derivative is $(a - b)f'(z(t))$. Multiplication by $\bar{a} - \bar{b}$ gives

$$(\bar{a} - \bar{b})(f(a) - f(b)) = |a - b|^2 \int_0^1 f'(b + t(a - b)) dt. \quad (6.1)$$

Rotation by $e^{i\chi}$ and taking imaginary parts proves the claim. $\square$

Only the polynomial $f(z) = z^m$ is used below. In that case the integrand is simply $m(b + t(a - b))^{m-1}$, and the identity requires no integration beyond the fundamental theorem of calculus for this polynomial in t .

Corollary 6.3 (Strict mediant refinement). *Suppose $a/b < c/d$ are consecutive in $F_{n-1} \cap [0, 1/2]$ and $b + d = n \geq 4$. On either new open interval formed by $(a + c)/n$, the new scalar root is strictly larger than the old one.*

Proof. Reflection permits $b < d$. Put

$$m = \lfloor (n - 1)/b \rfloor, \quad A = 2\pi(bx - a), \quad B = 2\pi(c - dx)/m.$$

The old scalar root $\rho \in (0, 1)$ satisfies

$$\rho^{d/m} \sin A + \rho^b \sin B = \sin(A + B), \quad A, B > 0, \quad A + B < \pi. \quad (6.2)$$

The mediant splits the interval according to the sign of

$$mB - A = 2\pi((a + c) - (b + d)x).$$

In the left child $A < mB$, keep count m initially; the smaller denominator remains b , and its two angles are $A, B - A/m$. In the right child $A > mB$, reflection puts the smaller denominator d first. Its denominators are $d, b + d$, its count is one because $b < d$, and its angles are $mB, A - mB$. The angle sums are respectively $A + B - A/m < \pi$ and $A < \pi$, so both scalar equations have unique roots. Their deficits at the old radius are

$$\begin{aligned} D_L &= \sin(A + B - A/m) - \rho^{(b+d)/m} \sin A - \rho^b \sin(B - A/m), \\ D_R &= \sin A - \rho^{b+d} \sin(mB) - \rho^d \sin(A - mB). \end{aligned}$$

Proving these deficits positive is enough: each new scalar left side increases strictly with the radius. To find their signs, we use the old equation to express each deficit using powers of a single radial variable. The left child combines exponents in units of $1/m$, whereas the right child has count one; this dictates the substitutions

$$\eta = A/m, \quad u = \rho^{b/m}, \quad v = \rho^{d/m}.$$

Set $G_m(a, b) = (\bar{a} - \bar{b})(a^m - b^m)$. Then the old scalar equation reads $v \sin A + u^m \sin B = \sin(A + B)$. In D_L , the mixed power is $\rho^{(b+d)/m} = uv$. Multiplying the old equation by u replaces $uv \sin A$ by $u \sin(A + B) - u^{m+1} \sin B$. This eliminates v and gives

$$D_L = \sin(B + (m - 1)\eta) - u \sin(B + m\eta) - u^m \sin(B - \eta) + u^{m+1} \sin B.$$

These four terms are the imaginary part of $e^{iB}(e^{-i\eta} - u)(e^{im\eta} - u^m)$. For D_R , both radial terms contain v^m , since $\rho^{b+d} = u^m v^m$ and $\rho^d = v^m$. We therefore eliminate u^m instead, using $u^m \sin B = \sin(A + B) - v \sin A$. After multiplication by the positive quantity $\sin B / \sin A$, the identity

$$\sin(A + B) \sin(mB) + \sin B \sin(A - mB) = \sin A \sin((m + 1)B)$$

then gives

$$\frac{\sin B}{\sin A} D_R = \sin B - v^m \sin((m + 1)B) + v^{m+1} \sin(mB).$$

This is the imaginary part of $e^{iB}(1 - ve^{-iB})(1 - v^m e^{imB})$; the remaining term $-v$ is real. Thus the two deficits are instances of the same divided difference:

$$\begin{aligned} D_L &= \operatorname{Im}\left(e^{iB} G_m(e^{i\eta}, u)\right), \\ D_R &= \frac{\sin A}{\sin B} \operatorname{Im}\left(e^{iB} G_m(1, ve^{iB})\right). \end{aligned} \tag{6.3}$$

The factors $\bar{a} - \bar{b}$ in these expressions record the directions of the two segments. Lemma 6.2 tests their signs by averaging the derivatives of $f(z) = z^m$, rotated through B . The left segment joins the positive real number u to $e^{i\eta}$; the right joins 1 to ve^{iB} . Their arguments lie respectively in $[0, \eta]$ and $[0, B]$. Since $\eta, B < \pi$, each segment lies in an open half-plane through the origin and avoids zero. For the left segment, the rotated derivative $e^{iB} m z^{m-1}$ has arguments in

$$[B, B + (m - 1)\eta] \subset (0, \pi), \quad B + (m - 1)\eta < A + B < \pi.$$

For the right segment its arguments lie in $[B, mB] \subset (0, \pi)$, because the right-child condition is $mB < A < \pi$. These statements also cover $m = 1$, when the derivative is constant. The lemma makes both imaginary parts positive; the factor $\sin A / \sin B$ is positive as well. Each child's scalar left side is therefore below its target at ρ , so its strictly increasing dependence on radius places the child root above ρ .

The right child's count is already correct. On the left, the actual count is $\lfloor (b + d)/b \rfloor$, whereas the comparison used $m = \lfloor (b + d - 1)/b \rfloor$. They differ precisely when $b \mid d$; coprimality of the two denominators makes this equivalent to $b = 1$. In that case $m = d$ and the actual count is $d + 1$. Lemma 6.1 increases the radius once more, starting with the valid angle sum $A + B - A/m < \pi$. $\square$

Theorem 6.4 (Monotonicity in the order). *For $n \geq 5$ and $0 \leq \theta \leq \pi$,*

$$K_{n-1}(\theta) \leq K_n(\theta).$$

At inherited Farey rays there is equality. At other rays equality holds precisely when both the containing Farey interval and its factor count remain unchanged.

Proof. Between consecutive fractions $a/b < c/d$ of F_{n-1} , either nothing is inserted in F_n , or the sole inserted fraction is $(a + c)/(b + d)$ with $b + d = n$. Indeed, if h/n is inserted, the unimodular coordinates

$$(n, h) = (cn - dh)(b, a) + (bh - an)(d, c)$$

have positive integer coefficients; since $b + d \geq n$, both coefficients must be one. On an unchanged interval the factor count stays fixed or increases by one, so the scalar equation is unchanged or Lemma 6.1 applies. A divided interval is covered by Corollary 6.3. An inherited endpoint has value one at both orders; a new endpoint had old radius below one and has new radius one. $\square$

6.1 Small orders and the boundary

Proposition 6.5. *One has $\Theta_1 = \{1\}$, $\Theta_2 = [-1, 1]$, and, with $\xi = e^{2\pi i/3}$,*

$$\Theta_3 = \text{conv}\{1, \xi, \bar{\xi}\} \cup [-1, -1/2]. \quad (6.4)$$

On $0 < \theta < \pi$, its radial maximum $R_3(\theta)$ is the order-three scalar root. Moreover $R_3(\theta) \leq K_4(\theta)$ for every $0 \leq \theta \leq \pi$.

Proof. An order-two stochastic matrix has eigenvalues 1 and $a - b$, where $a, b \in [0, 1]$; this proves the first two claims. If $x + iy$ is a nonreal eigenvalue of an order-three stochastic matrix, its spectrum is $1, x + iy, x - iy$. Nonnegative diagonal entries and off-diagonal products give

$$1 + 2x = \text{tr } A \geq 0, \quad (1 + 2x)^2 \leq 3 \sum_i a_{ii}^2 \leq 3 \text{tr } A^2 = 3(1 + 2x^2 - 2y^2).$$

Thus $x \geq -1/2$ and $3y^2 \leq (1 - x)^2$, which, with $x \leq 1$, places the eigenvalue in the displayed triangle. Conversely every point $a + b\xi + c\bar{\xi}$ of the triangle is an eigenvalue of $aI + bC_3 + cC_3^2$, where C_3 is a three-cycle permutation matrix, $a, b, c \geq 0$, and $a + b + c = 1$. The additional real interval is realized already at order two.

For the interval $(0, 1/3)$, $q = 1$ and $s = m = 3$; (5.9) gives the segment $\omega = \beta + \alpha\xi$. For $(1/3, 1/2)$, reflected denominator ordering gives $q = 2, s = 3, m = 1$, and the reduced equation factors as

$$\omega^3 - \beta\omega - \alpha = (\omega - 1)(\omega^2 + \omega + \alpha) = 0.$$

Its nonreal roots have real part $-1/2$ and trace the vertical side. These are the scalar roots by (5.8) and uniqueness. In particular $\lim_{\theta \uparrow \pi} R_3(\theta) = 1/2$, although $R_3(\pi) = 1$. To compare orders three and four, insert $1/4$ into $(0, 1/3)$ and apply Corollary 6.3; on $(1/3, 1/2)$ the interval stays fixed and the count increases from one to two, so Lemma 6.1 applies. The angle sums are strictly below π on these open intervals by (5.2). Endpoints, including the exceptional negative real ray, have the claimed comparison directly. $\square$

The unit-circle classification is also elementary. Cycles realize every root of unity of order at most n . Conversely, if $Ax = \lambda x$ with $|\lambda| = 1$, choose $M = \max_j |x_j| > 0$ and $S = \{j : |x_j| = M\}$. Equality in the triangle inequality shows, for every $i \in S$, that each positive transition $i \rightarrow j$ stays in S and satisfies $x_j = \lambda x_i$. There is a directed cycle in S of length $\ell \leq n$, and following it gives $\lambda^\ell = 1$. Thus the unit-circle points are exactly those asserted in the boundary theorem.

Completion of Theorem 1.1. Fix $n \geq 4$ and a ray $0 \leq \theta \leq \pi$, and let $R_n(\theta)$ be its actual radial maximum. Attainment gives $K_n(\theta) \leq R_n(\theta)$. On a Farey ray both sides equal one by the unit-circle classification. Otherwise put $z = R_n(\theta)e^{i\theta}$ and let $k \leq n$ be its least order of occurrence. Then $z \neq 0$, z is nonreal, $|z| < 1$, and $k \geq 3$. If $k = 3$, Proposition 6.5 and order monotonicity give

$$|z| \leq R_3(\theta) \leq K_4(\theta) \leq K_n(\theta) \leq |z|.$$

If $k \geq 4$, the invariant-polygon criterion says that its multiplier first admits a polygon at k vertices. An outward enlargement admitting at most k vertices would contradict its radial maximality in Θ_n . Thus Theorem 4.1 applies. The convex comparison and attainment give $|z| = K_k(\theta)$; then

$$|z| = K_k(\theta) \leq K_n(\theta) \leq |z|.$$

Hence $R_n = K_n$ on every ray. Star-shapedness gives

$$\Theta_n = \{re^{i\theta} : 0 \leq r \leq K_n(\theta)\},$$

with K_n extended by conjugation. Since this radial function is positive and continuous, its graph is exactly the boundary. Equations (5.9) and (5.8) identify its arcs with the stated Ito equations, and the endpoint limits join the prescribed roots of unity. Proposition 6.5 supplies the remaining orders and the terminal order-three segment. $\square$

7 Optimal polygonal gauges

Invariant polyhedra and their Minkowski functionals are standard tools in stability and control [10, 15, 11]. Hennet and Lasserre give the invariance condition for a rotation-dilation on a centered regular polygon [18, Lemma 2]; regular polygons also appear in constructions of polyhedral Lyapunov functions for planar complex modes [27]. Kousoulidis and Forni study polyhedral Lyapunov functions with prescribed complexity [22]. Here the completed boundary theorem determines the exact best contraction factor among all polygons with a prescribed vertex bound, allowing nonregular and nonsymmetric polygons.

Let R_θ be planar rotation and $T_{\rho,\theta} = \rho R_\theta$, with $\rho > 0$. For $N \geq 4$, define

$$\Gamma_N(T_{\rho,\theta}) = \inf\{\gamma > 0 : T_{\rho,\theta}P \subseteq \gamma P, \quad 0 \in \text{int } P, \quad P \text{ has at most } N \text{ vertices}\},$$

where P ranges over full-dimensional polygons. No symmetry is imposed. Its Minkowski functional

$$V_P(x) = \inf\{a > 0 : x \in aP\}$$

is therefore a gauge, and need not be a norm. By Theorem 1.1, the exact radial function of Θ_N is

$$K_N(\theta) = \max\{r : re^{i\theta} \in \Theta_N\}.$$

Theorem 7.1 (Exact gauge complexity). *For $N \geq 4$, $\rho > 0$, and $0 \leq \theta \leq \pi$,*

$$\Gamma_N(T_{\rho,\theta}) = \frac{\rho}{K_N(\theta)}. \quad (7.1)$$

The infimum is attained. An attaining polygon satisfies

$$V_P(T_{\rho,\theta}x) \leq \Gamma_N(T_{\rho,\theta})V_P(x) \quad (x \in \mathbb{R}^2).$$

Moreover,

$$\min_{0 \leq \theta \leq \pi} K_N(\theta) = \cos(\pi/N), \quad \max_{0 \leq \theta \leq \pi} \frac{\Gamma_N(T_{\rho,\theta})}{\rho} = \sec(\pi/N), \quad (7.2)$$

and the first Farey arc has the explicit radius

$$K_N(\theta) = \frac{\cos(\pi/N)}{\cos(\theta - \pi/N)} \quad (0 \leq \theta \leq 2\pi/N). \quad (7.3)$$

In particular the minimum in (7.2) is attained at $\theta = \pi/N$.

Proof. For every admissible polygon and every $\gamma > 0$,

$$\rho R_\theta P \subseteq \gamma P \iff (\rho/\gamma)R_\theta P \subseteq P.$$

The invariant-polygon criterion (2.1) puts $(\rho/\gamma)e^{i\theta}$ in Θ_N , so $\rho/\gamma \leq K_N(\theta)$. This proves the lower bound in (7.1).

Conversely, the boundary multiplier $K_N(\theta)e^{i\theta}$ belongs to Θ_N . If it is nonreal and has modulus less than one, the criterion supplies a polygon with at most N vertices; the observation following (2.1) makes it full dimensional with zero in its interior. At a nonreal unit-circle point, use the convex hull of its root-of-unity orbit, whose order is at most N . On the two

real rays, where $K_N = 1$, use a centered square. In each case $K_N(\theta)R_\theta P \subseteq P$. Rescaling this inclusion gives attainment and the reverse bound. Applying it to $x \in aP$ and taking the infimum over a gives the stated gauge inequality.

To find the worst angle, let Q_N be the convex hull of the N th roots of unity. For $z \in Q_N$, multiplication of a convex expression for z by any vertex of Q_N again gives a convex combination of the same roots. Thus $zQ_N \subseteq Q_N$, and the polygon criterion gives $Q_N \subseteq \Theta_N$. Since the inradius of Q_N is $\cos(\pi/N)$,

$$K_N(\theta) \geq \cos(\pi/N) \quad (0 \leq \theta \leq \pi).$$

For the first Farey interval, from $0/1$ to $1/N$, the scalar equation (5.3) has $q = 1$, $s = m = N$, $A = \theta$, and $B = 2\pi/N - \theta$. It is linear in its radial variable, and gives

$$K_N(\theta) = \frac{\sin(2\pi/N)}{\sin \theta + \sin(2\pi/N - \theta)} = \frac{\cos(\pi/N)}{\cos(\theta - \pi/N)}.$$

Evaluation at $\theta = \pi/N$ proves the minimum; the gauge formula then proves the maximum. $\square$

In the plane a polygon has as many sides as vertices. Thus N also bounds the number of linear pieces in the representation of V_P as a maximum of linear functionals. A strictly contracting gauge with this bound exists exactly when

$$\rho < K_N(\theta).$$

If the polygon may be chosen after the angle is known, but the same vertex bound must work for every angle, the exact condition is $\rho < \cos(\pi/N)$. The restriction $N \geq 4$ matters on the negative real ray: although $-1 \in \Theta_3$, an inclusion $-P \subseteq P$ implies $P = -P$, and a full-dimensional triangle cannot be centrally symmetric.

8 Farey asymptotics and vertex complexity

The preceding radius and contraction formulas are exact at every order. We now approximate their values as N increases. These estimates are consequences of the completed boundary theorem; they introduce no error into its description of Θ_N .

The arithmetic viewpoint has a separate history. Đoković used cyclic polygons generated by powers of a complex number in his simplification of the stochastic eigenvalue region [24]. Dubuc and Malik developed the connection among these convex hulls, trinomial equations, and Farey sequences [13]; Dubuc and Zaoui subsequently studied the fractal dimension of unions of trinomial arcs [14]. In particular, for a trinomial Farey arc associated with $p/q < r/s$ and $s > q > 2$, Dubuc and Malik prove the one-sided estimate $|z|^s \geq 1 - \pi^2/(qs)$ [13, Theorem 13.1]. The prescribed-ray scalar characterization of Kirkland, Laffey, and Šmigoc [20] gives a later point of comparison. Here the matrix order varies: the scalar equation yields a relative asymptotic that records both Farey denominators and the position within the interval. It refines the one-sided estimate in the trinomial subcase and also applies when the factor count is larger.

For an angle strictly between two order- N Farey rays, orient the interval as before so that its smaller endpoint denominator comes first:

$$\frac{p}{q} < y < \frac{r}{s}, \quad q < s, \quad rq - ps = 1, \quad m = \lfloor N/q \rfloor.$$

Here $y = \theta/(2\pi)$ or $1 - \theta/(2\pi)$ according to the reflection. Its relative coordinate is

$$t = s(qy - p) \in (0, 1), \quad y - \frac{p}{q} = \frac{t}{qs}, \quad \frac{r}{s} - y = \frac{1 - t}{qs}.$$

The coordinate t is allowed to approach either endpoint as rapidly as the Farey data permit.

Theorem 8.1 (Uniform radial deficit). *Uniformly over all open Farey intervals of order N and all $0 < t < 1$,*

$$1 - K_N(\theta) = \frac{2\pi^2 t(1-t)}{qs} \left(\frac{t}{s} + \frac{1-t}{mq} \right) (1 + O(N^{-2})) \quad (N \rightarrow \infty). \quad (8.1)$$

The implied constant is absolute. Thus, for a fixed irrational $x \in (0, 1/2)$, let $p_N/q_N < y_N < r_N/s_N$ be its successive oriented Farey intervals, with $y_N = x$ or $1-x$, and set $m_N = \lfloor N/q_N \rfloor$, $t_N = s_N(q_N y_N - p_N)$. Then

$$1 - K_N(2\pi x) \sim \frac{2\pi^2 t_N(1-t_N)}{q_N s_N} \left(\frac{t_N}{s_N} + \frac{1-t_N}{m_N q_N} \right),$$

without a lower bound on t_N or $1-t_N$.

Proof. Farey adjacency gives $q + s > N$. It follows that $s > N/2$; also $mq > N/2$, because $N < (m+1)q$ and $m \geq 1$. Set

$$A = \frac{2\pi t}{s}, \quad B = \frac{2\pi(1-t)}{mq}, \quad H(u) = u^{s/m} \sin A + u^q \sin B.$$

Both angles and their sum are $O(N^{-1})$, uniformly in the interval and in t . With $K = K_N(\theta)$, the exact scalar equation is $H(K) = \sin(A+B)$. Its defect at radius one factors as

$$\begin{aligned} D := H(1) - \sin(A+B) &= 4 \sin(A/2) \sin(B/2) \sin((A+B)/2) \\ &= \frac{AB(A+B)}{2} (1 + O(N^{-2})). \end{aligned} \quad (8.2)$$

The last estimate is relative, even near an endpoint. Dividing each sine by its own positive argument gives three factors of the form $1 + O(N^{-2})$; no division by t or $1-t$ is used in the error bound. Similarly,

$$\begin{aligned} H'(1) &= \frac{s}{m} \sin A + q \sin B \\ &= \frac{2\pi}{m} \left(t \frac{\sin A}{A} + (1-t) \frac{\sin B}{B} \right) = \frac{2\pi}{m} (1 + O(N^{-2})). \end{aligned} \quad (8.3)$$

Here the two weights sum to one, so this estimate also stays uniform as either weight tends to zero.

The mean-value theorem gives

$$D = (1-K)H'(\xi) \quad (K < \xi < 1).$$

It remains to replace $H'(\xi)$ by $H'(1)$ accurately enough. We do this in two steps. First, the regular-polygon bound in Theorem 7.1 gives $1-K = O(N^{-2})$, hence $|\log \xi| = O(N^{-2})$. Since both exponents $q, s/m$ are at most N , each power ξ^{q-1} and $\xi^{s/m-1}$ differs from one by $O(N^{-1})$. The two derivative terms are positive, so

$$\frac{H'(\xi)}{H'(1)} = 1 + O(N^{-1}).$$

Together with $D = O(N^{-3})$ and (8.3), this improves the initial deficit estimate to

$$1 - K = O(mN^{-3}) = O((qN^2)^{-1}).$$

For the second step use the sharper exponent bound $s/m \leq N/m < 2q$. Now $|\log \xi| = O((qN^2)^{-1})$, so both powers in the derivative differ from one by $O(N^{-2})$. Positivity again gives

$$\frac{H'(\xi)}{H'(1)} = 1 + O(N^{-2}).$$

Combining the mean-value identity with (8.2)–(8.3) yields

$$1 - K = \frac{mAB(A+B)}{4\pi}(1 + O(N^{-2})).$$

Substitution of the two angles proves (8.1). All constants used above are independent of the interval and of t , which also proves the stated fixed-irrational asymptotic. $\square$

Thus the displayed leading term approximates the positive deficit $1 - K_N$, with relative error bounded by a constant times N^{-2} . At a Farey endpoint the deficit and leading term are both exactly zero; the relative assertion concerns the open interval. For a general irrational angle, the denominators and the position inside its current interval both matter. Badly approximable angles give a simpler rate. Recall that x is badly approximable if some $c_x > 0$ satisfies

$$\left| x - \frac{a}{b} \right| \geq \frac{c_x}{b^2} \quad \text{for every rational } a/b.$$

We write $a_N \asymp_x b_N$ when their ratio is bounded above and below by positive constants depending only on x .

Corollary 8.2 (Contraction loss and vertex budgets). *Let $x \in [0, 1/2]$ and $N \geq 4$. If $x = a/b$ is rational in lowest terms, $K_N(2\pi x) = 1$ exactly when $b \leq N$. For irrational x , the radius is strictly below one at every finite order. For badly approximable x ,*

$$1 - K_N(2\pi x) \asymp_x N^{-3}, \quad \frac{\Gamma_N(T_{\rho, 2\pi x})}{\rho} - 1 \asymp_x N^{-3}. \quad (8.4)$$

Uniformly in angle,

$$\max_{0 \leq \theta \leq \pi} \left(\frac{\Gamma_N(T_{\rho, \theta})}{\rho} - 1 \right) = \sec(\pi/N) - 1 \sim \frac{\pi^2}{2N^2}. \quad (8.5)$$

For $\varepsilon > 0$, define

$$B_x(\varepsilon) = \min \left\{ N \geq 4 : \frac{\Gamma_N(T_{\rho, 2\pi x})}{\rho} - 1 \leq \varepsilon \right\}.$$

Then $B_x(\varepsilon) \asymp_x \varepsilon^{-1/3}$ for badly approximable x as $\varepsilon \downarrow 0$. If the same accuracy must hold for every angle, the required budget is exactly

$$B_{\text{unif}}(\varepsilon) = \max \left\{ 4, \left\lceil \frac{\pi}{\arccos((1 + \varepsilon)^{-1})} \right\rceil \right\} \sim \frac{\pi}{\sqrt{2\varepsilon}}. \quad (8.6)$$

These relative-loss budgets do not depend on $\rho > 0$.

Proof. The rational and irrational statements follow from the unit-circle classification in Theorem 1.1. Bad approximability is preserved by replacing x with $1 - x$. For the oriented Farey interval its two endpoint inequalities give

$$t \geq c_x s/q, \quad 1 - t \geq c_x q/s.$$

Since $q < s$ and $t < 1$, the first inequality yields $c_x s < q < s$. Thus the denominators are comparable. It also gives $t \geq c_x$, and the second inequality now gives $1 - t > c_x^2$. Together with $q + s > N$ and $s \leq N$, this proves

$$q \asymp_x s \asymp_x N, \quad mq \asymp N.$$

Every factor in the leading term of (8.1) therefore has the stated scale: the denominator qs is of order N^2 , while the parenthesis is of order N^{-1} . This proves the first estimate in (8.4). The second follows from

$$\frac{\Gamma_N(T_{\rho, 2\pi x})}{\rho} - 1 = \frac{1 - K_N(2\pi x)}{K_N(2\pi x)}$$

and $K_N \rightarrow 1$.

Equation (7.2) gives the exact maximum in (8.5); the Taylor expansion of the secant gives its asymptotic. The finite-order losses at an irrational angle are positive, so $B_x(\varepsilon) \rightarrow \infty$ as $\varepsilon \downarrow 0$. Inverting the two-sided N^{-3} bound then gives upper and lower constant multiples of $\varepsilon^{-1/3}$. Finally, solving $\sec(\pi/N) - 1 \leq \varepsilon$ for the integer $N \geq 4$ gives (8.6), and the expansion $\arccos((1 + \varepsilon)^{-1}) \sim \sqrt{2\varepsilon}$ gives its last term. $\square$

A Quantitative and unequal-exponent comparisons

The convex comparison contains more information than the sharp radius. Keeping its Jensen gap measures how far an admissible factor tuple is from equal coefficients. Allowing different exponents leads to a related optimization, in which the available argument range of each factor becomes an upper constraint.

A.1 Quantitative rigidity on the prescribed phase

Proposition A.1 (Quantitative rigidity). *Under the hypotheses of Theorem 5.2, put*

$$F(u) = \log \sin M - \log \sin(M - u), \quad \bar{u} = A + B, \quad J = \sum_{j=1}^m F(u_j) - mF(\bar{u}),$$

and let

$$D_\rho = \sin(A + B) - \rho^{s/m} \sin A - \rho^q \sin B.$$

Then

$$J \geq \frac{1}{2} \sum_{j=1}^m (u_j - \bar{u})^2, \quad D_\rho = \rho^{s/m} \sin A (e^{J/m} - 1). \quad (\text{A.1})$$

In particular,

$$\sum_{j=1}^m (u_j - \bar{u})^2 \leq 2m \log \left(1 + \frac{D_\rho}{\rho^{s/m} \sin A} \right) \leq \frac{2m D_\rho}{\rho^{s/m} \sin A}. \quad (\text{A.2})$$

With $\bar{\beta} = m^{-1} \sum_j \beta_j$ and $c_\rho = \rho^q \sin A / (1 + \rho^q)^2$, one also has

$$\sum_{j=1}^m (\beta_j - \bar{\beta})^2 \leq c_\rho^{-2} \sum_{j=1}^m (u_j - \bar{u})^2. \quad (\text{A.3})$$

For fixed interior Farey data and angles, both variances are $O(\rho_* - \rho)$ as an admissible family approaches the scalar root ρ_* . The constants are uniform on compact subsets of an open Farey interval when ρ is bounded away from zero.

Proof. Since $(1 - \Re \omega^q) / \Im \omega^q = -\cot M$, the normalized-line function from Theorem 5.2 is

$$\mathcal{D}(u) = \cos u - \cot M \sin u = \frac{\sin(M - u)}{\sin M} = e^{-F(u)}.$$

The prescribed real phase gives $\sum_j u_j = m\bar{u}$. The curvature bound (5.6) gives $F'' \geq 1$, so Taylor's inequality at $\bar{u}$ gives

$$F(u_j) \geq F(\bar{u}) + F'(\bar{u})(u_j - \bar{u}) + \frac{1}{2}(u_j - \bar{u})^2.$$

The linear terms cancel on summation, proving the gap bound. For its relation to the radius, the modulus identity gives $J/m = (q - s/m) \log \rho + \log \mathcal{D}(A + B)$, so

$$e^{J/m} = \rho^{q-s/m} \mathcal{D}(A + B) = \frac{\sin(A + B) - \rho^q \sin B}{\rho^{s/m} \sin A},$$

where the last equality is the normalized-line evaluation used there. Rearranging proves the exact identity in (A.1); taking logarithms and using $\log(1 + x) \leq x$ proves (A.2).

To pass from arguments to coefficients, (5.6) yields

$$\frac{du}{d\beta} = \frac{\rho^q \sin A}{|\rho^q e^{iA} - \beta|^2} \geq \frac{\rho^q \sin A}{(1 + \rho^q)^2} = c_\rho > 0.$$

Thus $|\beta_i - \beta_j| \leq c_\rho^{-1}|u_i - u_j|$. Apply this to the pairwise variance identity

$$\sum_{j=1}^m (x_j - \bar{x})^2 = \frac{1}{m} \sum_{i < j} (x_i - x_j)^2$$

to obtain (A.3). Finally, for $H(t) = t^{s/m} \sin A + t^q \sin B$,

$$D_\rho = H(\rho_*) - H(\rho) = H'(\rho_*)(\rho_* - \rho) + o(\rho_* - \rho).$$

The stated order and compact uniformity follow from the positive denominators in the variance bounds and continuity of the displayed quantities. $\square$

Here admissibility includes both the finite product and its specified real phase lift. The conclusion controls this factor tuple; it makes no matrix-distance assertion for arbitrary eigenvalues near the boundary.

A.2 Different exponents and constrained angular allocation

With different exponents the normalized factors lie on different lines. Measuring each argument from its limiting direction M_j gives an angular deficit $v_j = M_j - u_j$. The available deficits U_j need not agree. The optimal choice shares the deficit equally among the remaining factors after each binding upper constraint is met.

Theorem A.2 (Unequal-exponent product bound). *Let $m \geq 1$, let $q_1, \dots, q_m$ be positive integers, let $e \in \mathbb{Z}$, and write $z = \rho e^{i\theta}$ with $\rho > 0$. Suppose $A_j = \text{Arg}(z^{q_j}) \in (0, \pi)$ and $0 \leq \beta_j < 1$. Put*

$$M_j = \text{Arg}(z^{q_j} - 1), \quad u_j = \text{Arg}(z^{q_j} - \beta_j), \quad U_j = M_j - A_j, \quad v_j = M_j - u_j,$$

and assume the product and its specified real phase

$$z^e \prod_{j=1}^m (z^{q_j} - \beta_j) = \prod_{j=1}^m (1 - \beta_j), \quad e\theta + \sum_{j=1}^m u_j = 2\pi L, \quad L \in \mathbb{Z}. \quad (\text{A.4})$$

Then $0 < v_j \leq U_j < \pi$ and

$$S := \sum_{j=1}^m M_j + e\theta - 2\pi L = \sum_{j=1}^m v_j, \quad 0 < S \leq \sum_{j=1}^m U_j.$$

If $S < \sum_j U_j$, let $c \in (0, \max_j U_j)$ be the unique solution of $\sum_j \min(c, U_j) = S$; at equality set $c = \max_j U_j$. The vector

$$v_j^* = \min(c, U_j)$$

is the unique maximizer of $\prod_j \sin v_j$ subject to $0 < v_j \leq U_j$ and $\sum_j v_j = S$. Consequently

$$\rho^e \prod_{j=1}^m \sin M_j = \prod_{j=1}^m \sin v_j \leq \prod_{j=1}^m \sin v_j^*, \quad (\text{A.5})$$

with equality precisely when $v_j = v_j^*$ for every j . More precisely,

$$\log \frac{\prod_j \sin v_j^*}{\rho^e \prod_j \sin M_j} \geq \frac{1}{2} \sum_{j=1}^m (v_j - v_j^*)^2. \quad (\text{A.6})$$

If $S/m \leq \min_j U_j$, this reduces to

$$\rho^e \prod_{j=1}^m \sin M_j \leq \sin^m(S/m),$$

with equality exactly when $v_1 = \dots = v_m = S/m$.

Proof. As β increases from zero towards one, $\text{Arg}(z^{q_j} - \beta)$ increases from A_j towards M_j . Thus $0 < v_j \leq U_j < \pi$. Put $w_j = z^{q_j}$ and $c_j = (1 - \Re w_j)/\Im w_j = -\cot M_j$; here c_j may have either sign. The normalized-line identity $\Re g_j + c_j \Im g_j = 1$, with $g_j = (w_j - \beta_j)/(1 - \beta_j)$, gives

$$\frac{|w_j - \beta_j|}{1 - \beta_j} = \frac{1}{\cos u_j + c_j \sin u_j} = \frac{\sin M_j}{\sin v_j}.$$

Taking moduli in (A.4) proves the equality in (A.5); the phase gives $S = \sum_j v_j$. The function $c \mapsto \sum_j \min(c, U_j)$ is continuous and strictly increasing until it reaches $\sum_j U_j$. This proves the existence and uniqueness of the stated c , and makes v^* feasible.

Set $\Phi(v) = \sum_j \log \sin v_j$. Its Hessian is diagonal with entries $-\csc^2 v_j \leq -1$, so every feasible vector satisfies

$$\Phi(v) - \Phi(v^*) \leq \sum_j \cot(v_j^*)(v_j - v_j^*) - \frac{1}{2} \sum_j (v_j - v_j^*)^2.$$

When $S < \sum_j U_j$, the equal coordinate sums cancel the common slope $\cot c$. The remaining linear term is

$$\sum_{U_j < c} (\cot U_j - \cot c)(v_j - U_j) \leq 0.$$

Each coefficient is positive because cotangent decreases on $(0, \pi)$, and each difference is nonpositive by its upper constraint. It follows that v^* uniquely maximizes Φ , with precisely the stated quadratic deficit. Exponentiation gives the sharp product bound and its equality case. At total saturation the feasible set is the singleton $v = U$. If $S/m \leq \min_j U_j$, the choice $c = S/m$ gives the final assertion. $\square$

For the equal-exponent data of Theorem 5.2, all $M_j = M$ and $U_j = M - A$, while $S = m(M - A - B) < m(M - A)$. No upper constraint binds, and equality gives $v_j = M - A - B$, hence $u_j = A + B$ and equal coefficients. With unequal exponents it is the angular deficits that agree where the constraints do not bind. The theorem is a product comparison on a prescribed phase; identifying a spectral boundary would still require a realization and an independent argument that its selected branch is outermost.

B Independent-parameter realizations

Theorem 5.3 gives the full characteristic polynomial of the graph in Figure 3. We now record its support and connectivity, its endpoint matrices, and the equality case on a prescribed Farey arc.

Proposition B.1 (Structure of the realization family). *Let $A(\beta)$ be the matrix constructed in Theorem 5.3, and suppose $q < s$. If every $\beta_j \in (0, 1)$, it is irreducible and has exactly $n_0 + m$ positive entries: m rows have two, and every other row has one. If also $\gcd(q, s) = 1$, it is primitive. For every parameter tuple in $[0, 1]^m$ and every $n \geq n_0$, $A(\beta) \oplus I_{n-n_0}$ is stochastic of order n and retains all its eigenvalues.*

Proof. Suppose all return and connection weights are positive. The connections reach every block, and the returns then reach every vertex of each block; inserted vertices also lie on a connection. The graph is therefore strongly connected. Any terminal lies on its block cycle of length q and on the connection cycle of length s . If $\gcd(q, s) = 1$, every integer $L \geq (q-1)s$ has the form $aq + bs$ with $a, b \geq 0$: choose $b \in \{0, \dots, q-1\}$ so that $bs \equiv L \pmod{q}$. Given any two vertices, take a path from the first to a fixed terminal and a path from there to the second. Inserting the two terminal cycles supplies every sufficiently large additional length.

There are finitely many vertex pairs, so every sufficiently high power of the matrix is entrywise positive. This proves primitivity directly.

The two destinations of each terminal row are distinct. For $m \geq 2$ they lie in different blocks or one is an inserted vertex; for $m = 1$, the assumption $q < s$ requires an inserted connection. Hence there are exactly m rows with two positive entries and $n_0 - m$ with one. The identity-block extension has characteristic polynomial $(t - 1)^{n-n_0} \det(tI - A)$. $\square$

For consecutive order- n Farey denominators $q < s$, take $m = \lfloor n/q \rfloor$. Then $q + s > n \geq mq$ verifies the realization hypothesis, $n_0 \leq n$, and Farey coprimality gives primitivity for interior weights. When every parameter equals β , writing $\alpha = 1 - \beta$ reduces the characteristic polynomial to

$$\begin{cases} (t^q - \beta)^m - \alpha^m t^{mq-s}, & s \leq mq, \\ t^{s-mq} (t^q - \beta)^m - \alpha^m, & s > mq. \end{cases}$$

At the two constant endpoint tuples this becomes, respectively,

$$\det(tI - A(\mathbf{1})) = t^{n_0-mq} (t^q - 1)^m, \quad \det(tI - A(\mathbf{0})) = t^{n_0-s} (t^s - 1).$$

Thus the endpoint roots of unity are realized in the same continuous matrix family. Formula (5.10) also remains valid at every mixed tuple of zero, unit, and interior weights; irreducibility is asserted only for interior tuples.

The equality statement for the scalar bound singles out one tuple within this larger family. It uses the prescribed real phase: the characteristic equation by itself records the phase only modulo 2π .

Corollary B.2 (Equality within the realization family). *Let $n \geq 4$, and let $p/q < r/s$, $q < s$, be the denominator-oriented form of two consecutive fractions in F_n^+ . Put $m = \lfloor n/q \rfloor$ and suppose $0 \leq \beta_j < 1$. Let $\omega = \rho e^{i\vartheta}$, $0 < \rho < 1$, be a root of (5.10), with*

$$\frac{p}{q} < \frac{\vartheta}{2\pi} < \frac{r}{s}, \quad A = q\vartheta - 2\pi p, \quad B = \frac{2\pi r - s\vartheta}{m}.$$

Set $M = \text{Arg}(\omega^q - 1)$ and $u_j = \text{Arg}(\omega^q - \beta_j) \in [A, M)$. If

$$(s - mq)\vartheta + \sum_{j=0}^{m-1} u_j = 2\pi(r - mp),$$

then, with $\theta = \min\{\vartheta, 2\pi - \vartheta\}$,

$$\rho \leq K_n(\theta).$$

Equality holds if and only if all parameters agree. At equality their common value is uniquely

$$\beta_* = \frac{K_n(\theta)^q \sin B}{\sin(A + B)}.$$

Proof. Since $\omega \neq 0$, its characteristic equation is equivalent to

$$\omega^{s-mq} \prod_j (\omega^q - \beta_j) = \prod_j (1 - \beta_j).$$

The Farey angle calculation (5.2) gives $A, B > 0$ and $A + B < \pi$. Together with the assumed real phase, this is exactly the input to Theorem 5.2. It gives the bound and equality precisely when the parameters agree. In that case their factor arguments all equal $A + B$; taking real and imaginary parts as in (5.8) gives the stated common value. Conversely, (5.7) supplies that tuple and realizes the scalar radius. $\square$

Along the selected open arc, β_* tends to one at the denominator- q endpoint and to zero at the denominator- s endpoint. Indeed, $K_n \rightarrow 1$ there, while respectively $A \rightarrow 0$ and $B \rightarrow 0$ in its formula. The other angle stays strictly between zero and π by (5.2). The matrices realizing the arc therefore converge to the two endpoint matrices displayed above.

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Use of generative artificial intelligence

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